

PRACTICAL 12

AIM : Sharing, Publishing, and Presenting Data Visualizations.

What you will learn ?

- a) Export dashboards as PDF/Image
- b) Use presentation mode in Tableau
- c) Publish dashboards to Tableau Public
- d) Share dashboards via Tableau Server/Online
- e) Enable user interaction on shared dashboards
- f) Manage permissions and access control
- g) Evaluate and optimize sharing strategies

Version Compatibility Note:

- Tableau Desktop (Free Trial / Student / Public 2026.2): Full native support for local exports (PDF, PNG, EMF), Presentation Mode, publishing to Tableau Public, and clientside interactions (Actions, Tooltip Vizizes).
- Tableau Server / Tableau Cloud Specific Features (Parts D & F): Direct enterprise publishing, project folder governance, and active directory user permissions require a paid Tableau Server/Cloud environment. We will execute local and Tableau Public equivalents while covering the exact enterprise steps.

Dataset: Netflix Movies and TV Shows

Kaggle link: <https://www.kaggle.com/datasets/shivamb/netflix-shows?resource=download>

A: Export Dashboards as PDF/Image

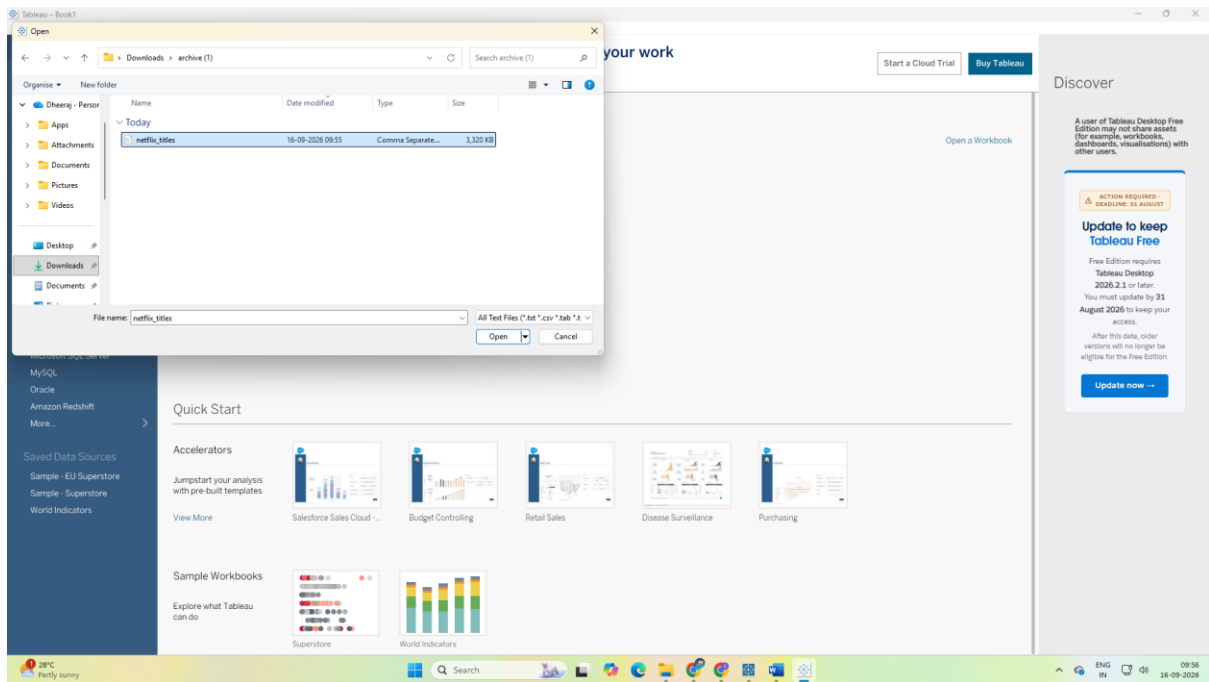
Objective: Build an executive Netflix overview dashboard and export high-resolution vector PDFs and raster images (.png / .emf) formatted for print briefings and slide decks.

Stakeholders and executives often require static offline artifacts (board decks, email reports). Knowing how to configure print setups, page scaling, and vector exports ensures charts do not become cropped, blurry, or misaligned when exported outside of Tableau.

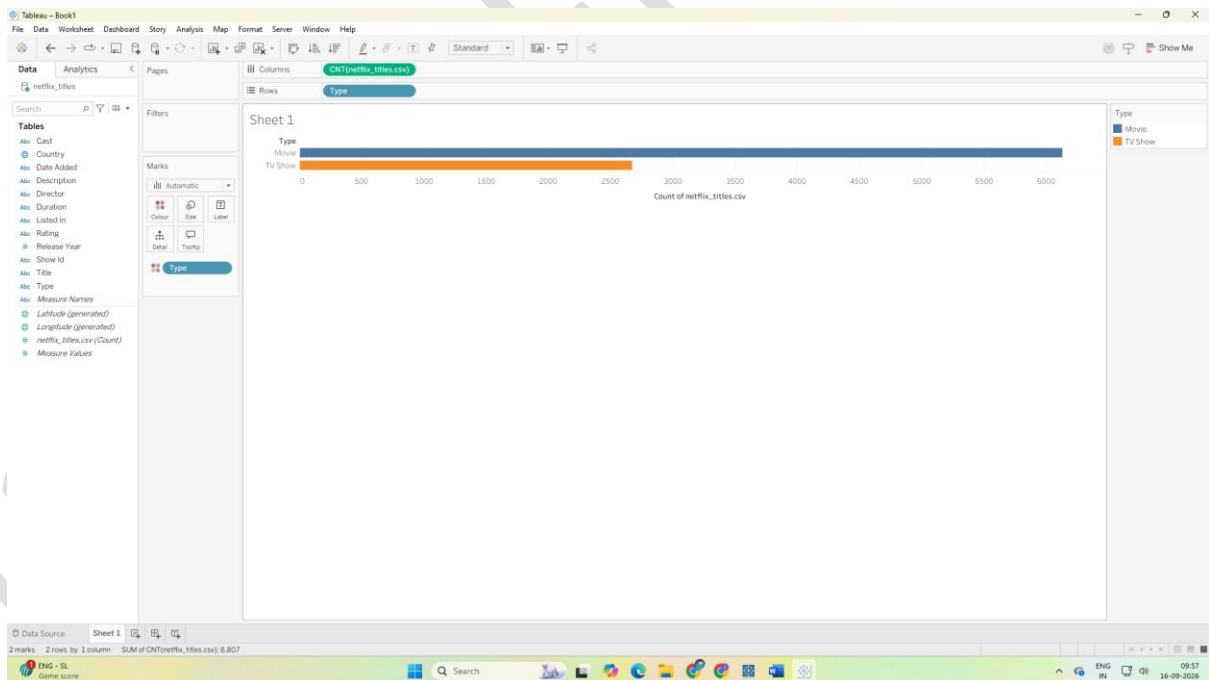
Step 1: Build Two Base Visualizations

Sheet 1: Content Type Split (Bar Chart)

1. Connect to netflix_titles.csv (use Text file connection).



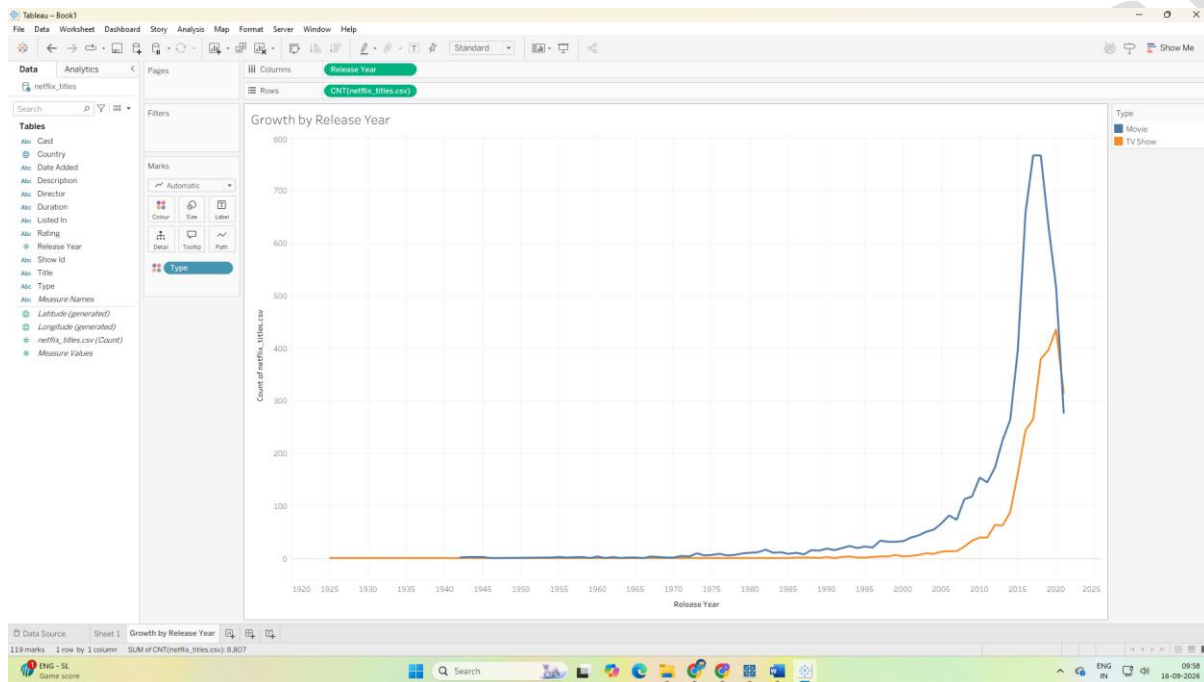
2. Drag type to Rows.
3. Drag netflix_titles.csv (Count) (or number of records) to Columns.
4. Drag type to Color (Marks card).



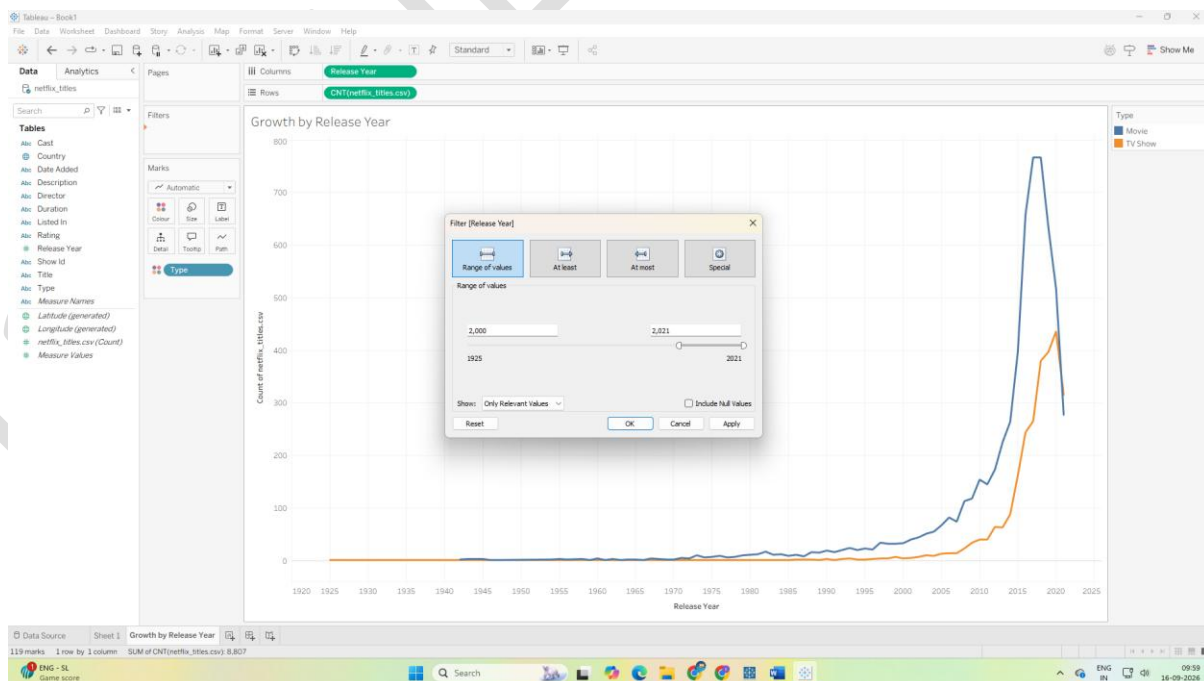
5. Result: A horizontal bar chart showing Movies vs. TV Shows.

Sheet 2: Release Year Growth (Line Chart)

1. Create a worksheet named: Growth by Release Year.
2. Drag release_year to Columns (ensure it is Continuous or an unbroken timeline).
3. Drag netflix_titles.csv (Count) to Rows.
4. Drag type to Color.



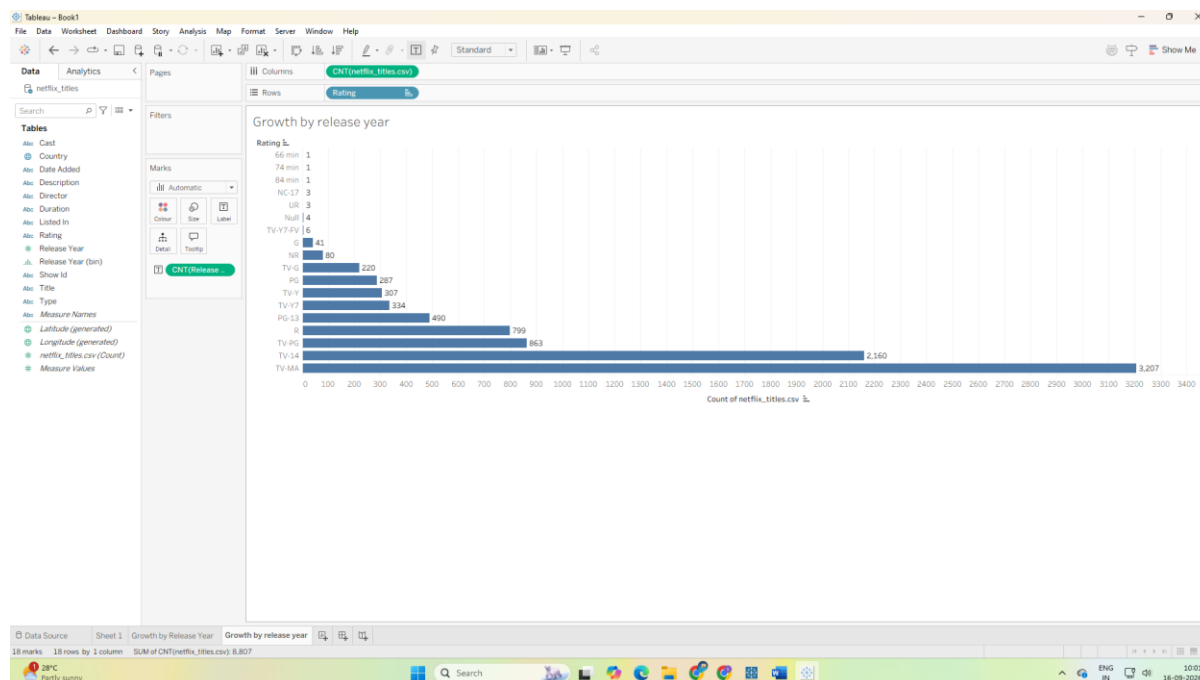
5. Filter: Drag release_year to Filters select Range of values set from 2000 to 2021(removes sparse historical outliers).



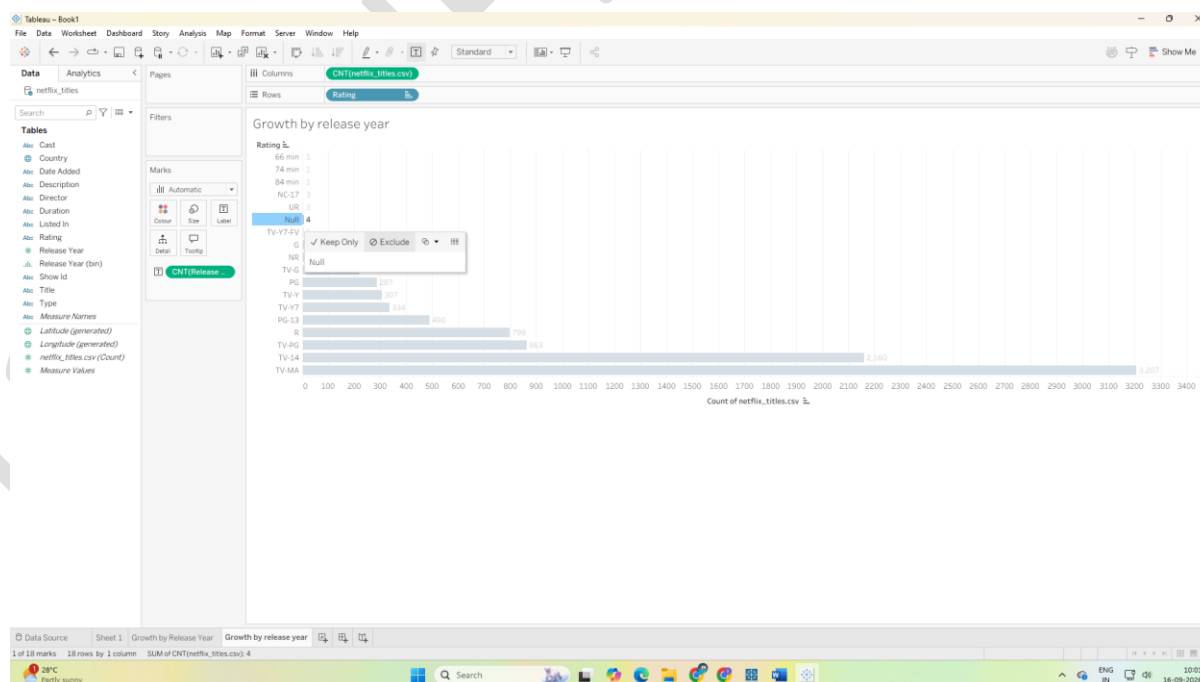
6. Result: A dual-line trend chart showing content volume over time.

Sheet 3: Top 10 Ratings

1. Create a worksheet named: Growth by Release Year.
2. Drag rating to Rows.
3. Drag netflix_titles.csv (Count) to Columns.



4. Sort descending by count (Click).
5. Filter out Null ratings if any appear.

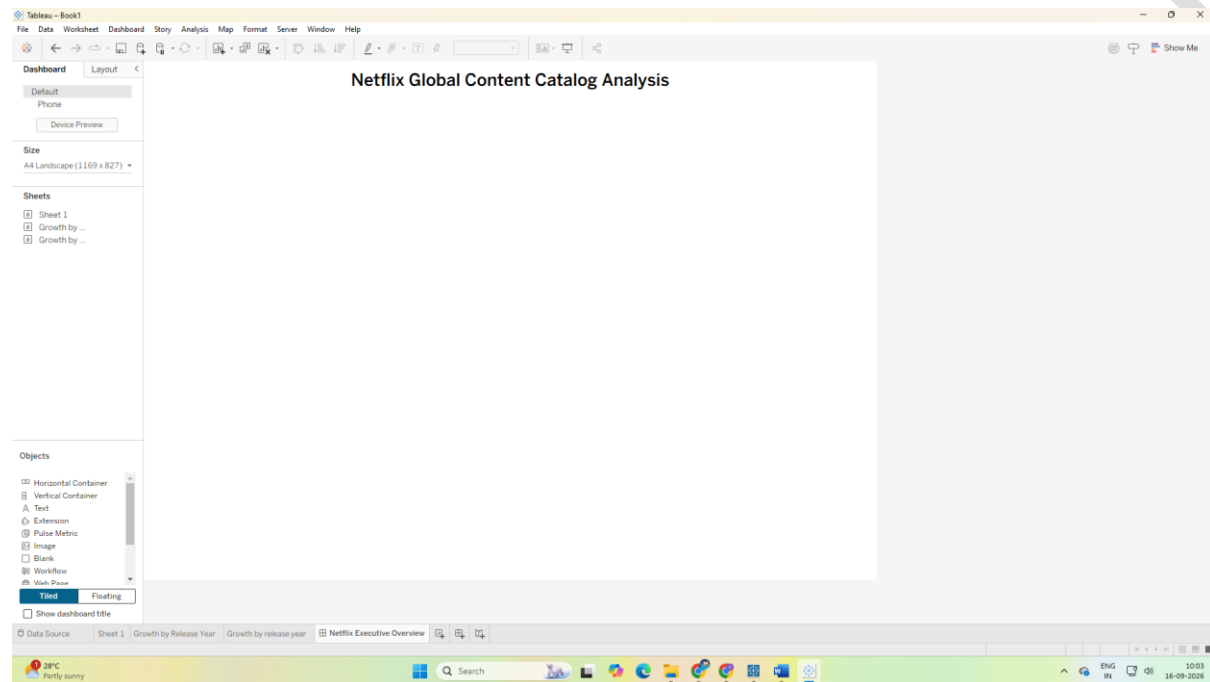


Step 2: Assemble the Executive Dashboard

1. Click the New Dashboard tab and rename it: Netflix Executive Overview.

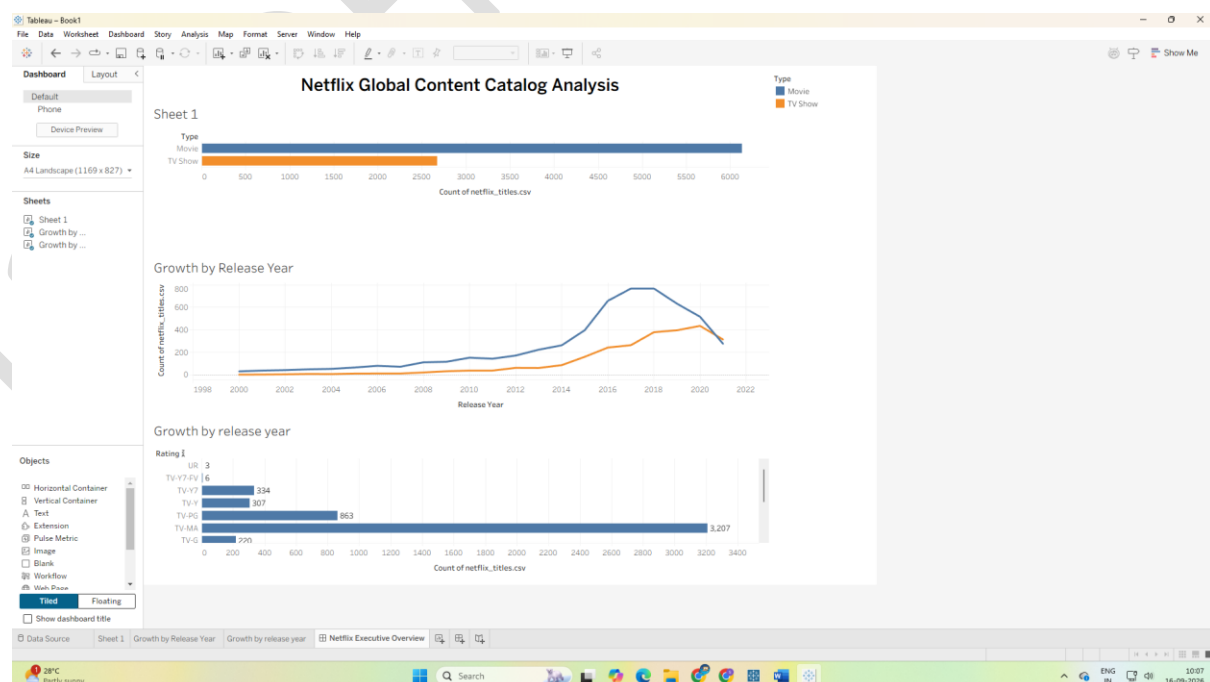
2. In the Dashboard pane on the left, under Size, select the dropdown:

o Choose Fixed Size --> A4 Landscape (1169 x 827 px) or Letter Landscape (1100 x 850 px). (Fixed size prevents layout shifts during PDF export).



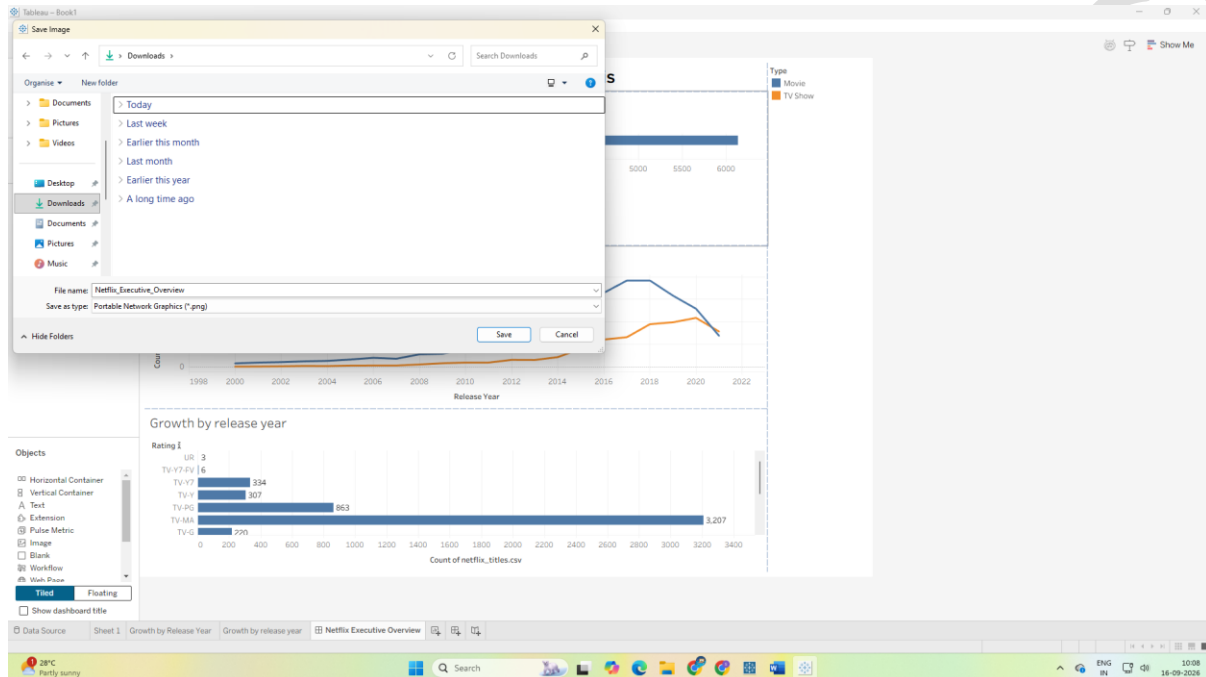
5. Drag your sheets into the canvas:

- Place Content Type Split on the top left.
- Place Top 10 Ratings on the top right.
- Place Growth by Release Year across the bottom.



Step 3: Export as High-Resolution Image (Raster / Vector)

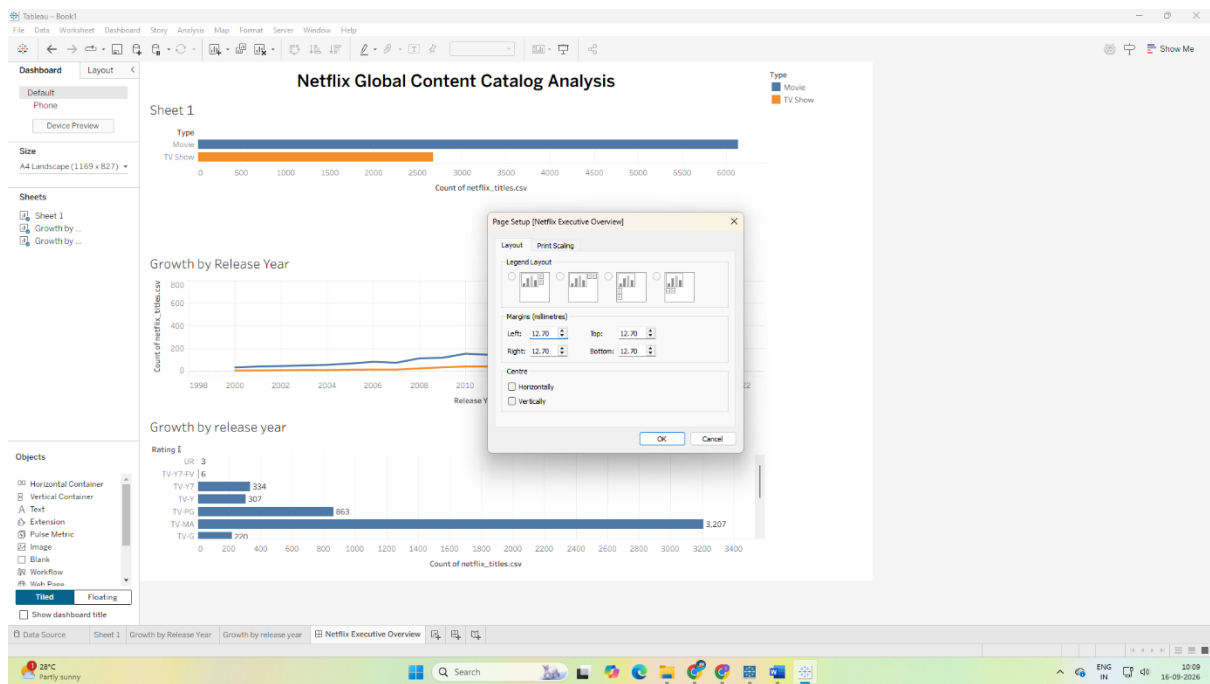
1. Navigate to the top application menu bar:
 - o Select Dashboard --> Export Image... (or Worksheet --> Export --> Image...).
2. In the file save dialogue:
 - o Choose file location and name it: Netflix_Executive_Overview.png.
 - o Save as type: PNG Image (*.png) or Enhanced Metafile (*.emf) for vector scaling in PowerPoint.



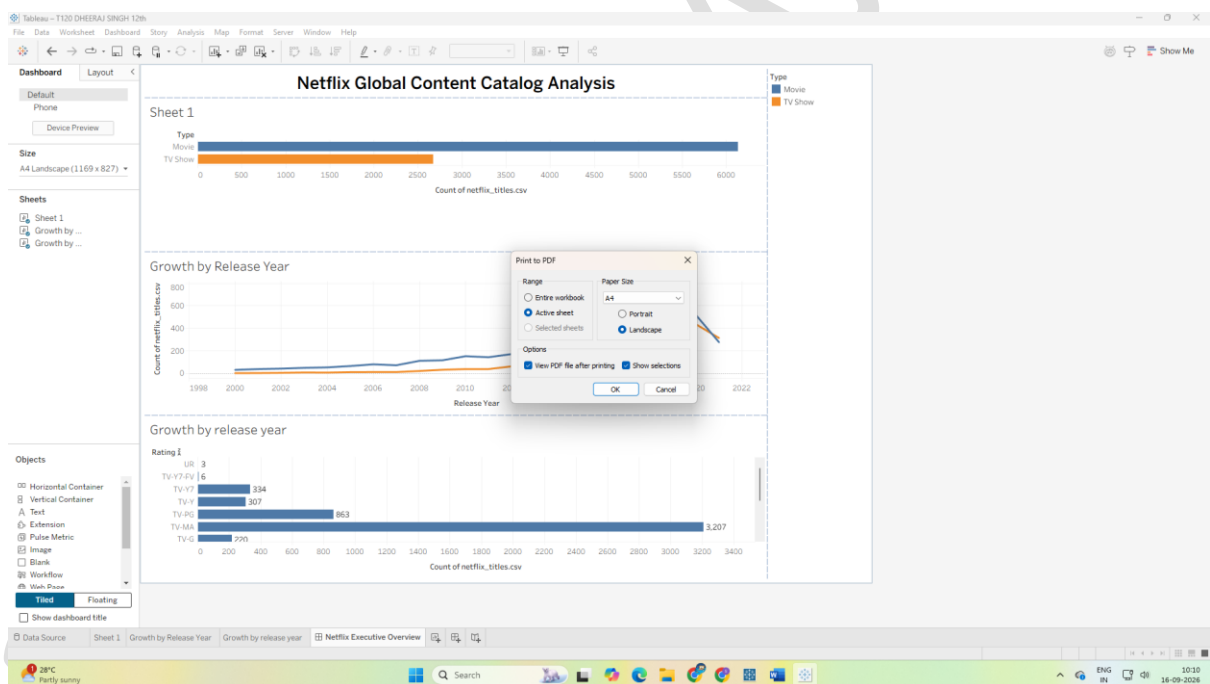
3. Click Save.

Step 4: Configure Print Setup and Export to PDF

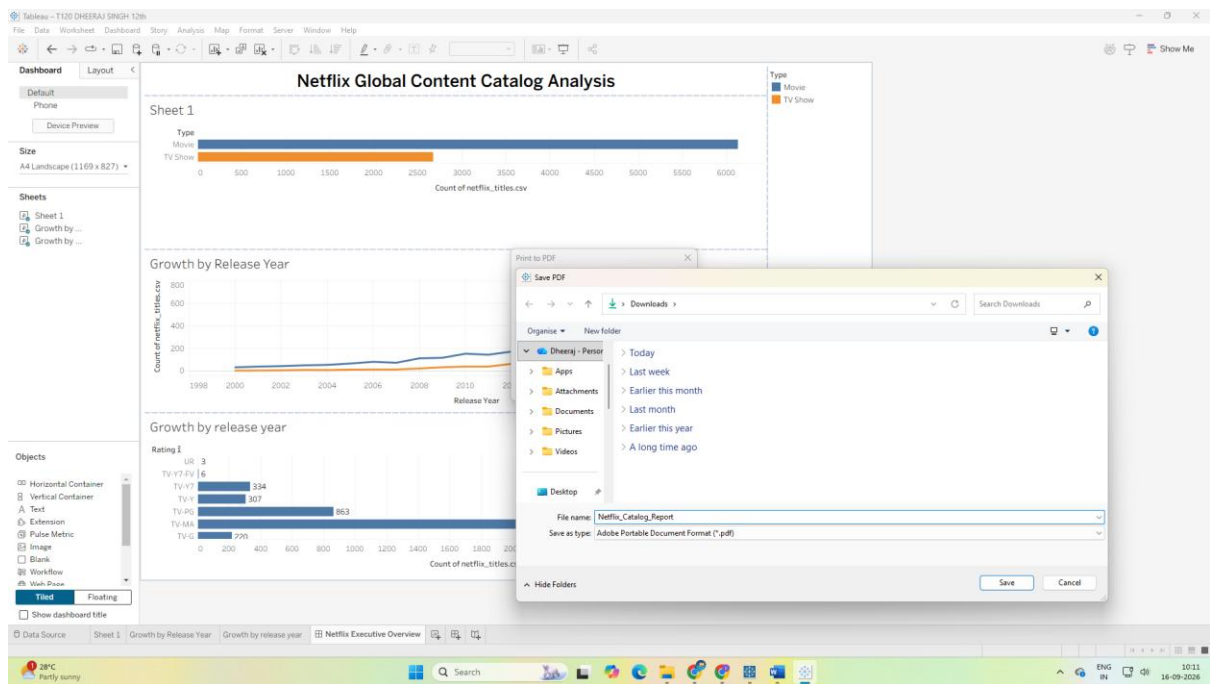
1. In the top menu, go to File --> Page Setup...:
 - o Page Type: Select A4 (or Letter).
 - o Orientation: Select Landscape.
 - o Scaling: Select Fit to at most 1 page wide by 1 page high (ensures no cutoff).
 - o Click OK.



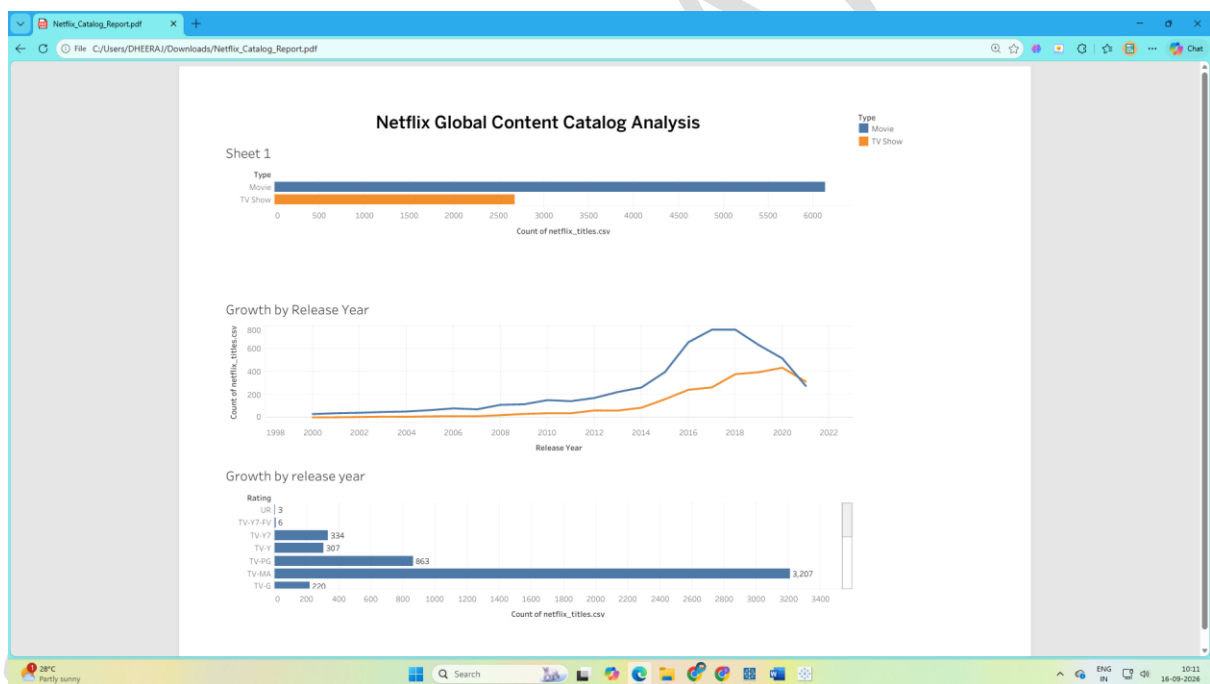
2. Go to File --> Print to PDF...:



3. Click OK, choose the destination folder, and name the file Netflix_Catalog_Report.pdf.



4. Open the generated PDF to confirm zero pixelation and complete margin alignment



B: Use Presentation Mode in Tableau

Compatibility Note : You need Tableau Desktop Public Edition (100% Free Forever).

Objective: Use Tableau's built-in Presentation Mode to display visualizations in a distraction-free, full-screen environment for live meetings and stakeholder presentations.

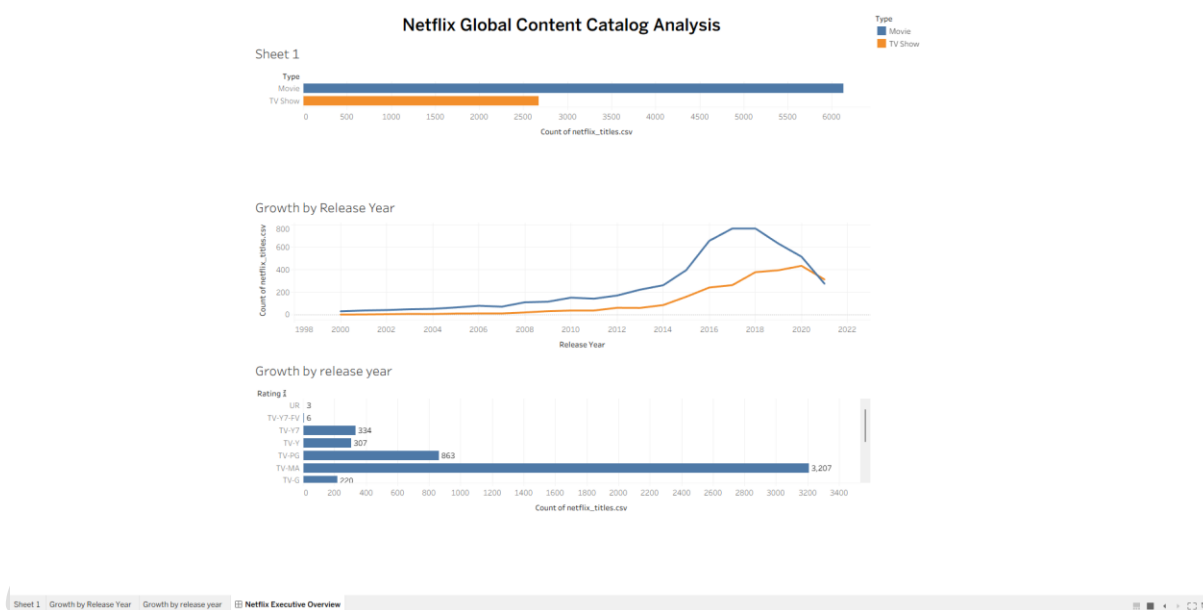
During executive briefings, application sidebars, cards, shelves, and window chrome introduce unnecessary clutter. Presentation mode hides the development interface, retains all interactive tooltips and cross-filtering, and provides a lightweight presentation toolbar for smooth navigation.

Step 1: Launch Presentation Mode

1. Open your Netflix Executive Overview dashboard in Tableau Desktop.

2. Enter Presentation Mode using any of these methods:

- o Keyboard Shortcut: Press F7 (Windows) or Cmd + Enter (Mac).
- o Toolbar Icon: Click the Presentation Mode icon (a small projection screen icon) in the top application toolbar.
- o Menu Path: Go to the top menu and select Window --> Presentation Mode.

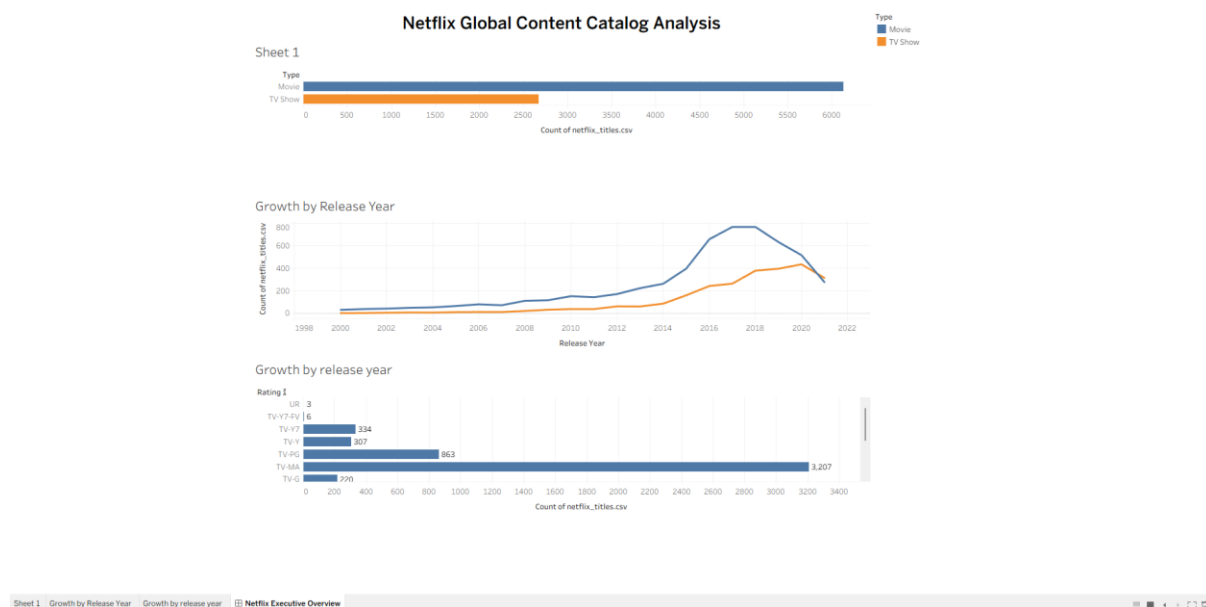


Step 2: Use On-Screen Presentation Controls

Once in full-screen mode, an unobtrusive toolbar appears at the bottom or top corner of the display:

- Sheet Navigation: Use the left/right arrows on the toolbar (or the Page Up / Page Down keys) to cycle through all worksheets, dashboards, and stories in your workbook.
- Live Interactive Filtering: Click on any bar or line mark (e.g., clicking Movies in the split chart) to demonstrate real-time cross-filtering.
- Interactive Tooltips: Hover over data points to present granular numbers on demand without cluttering the main slide.

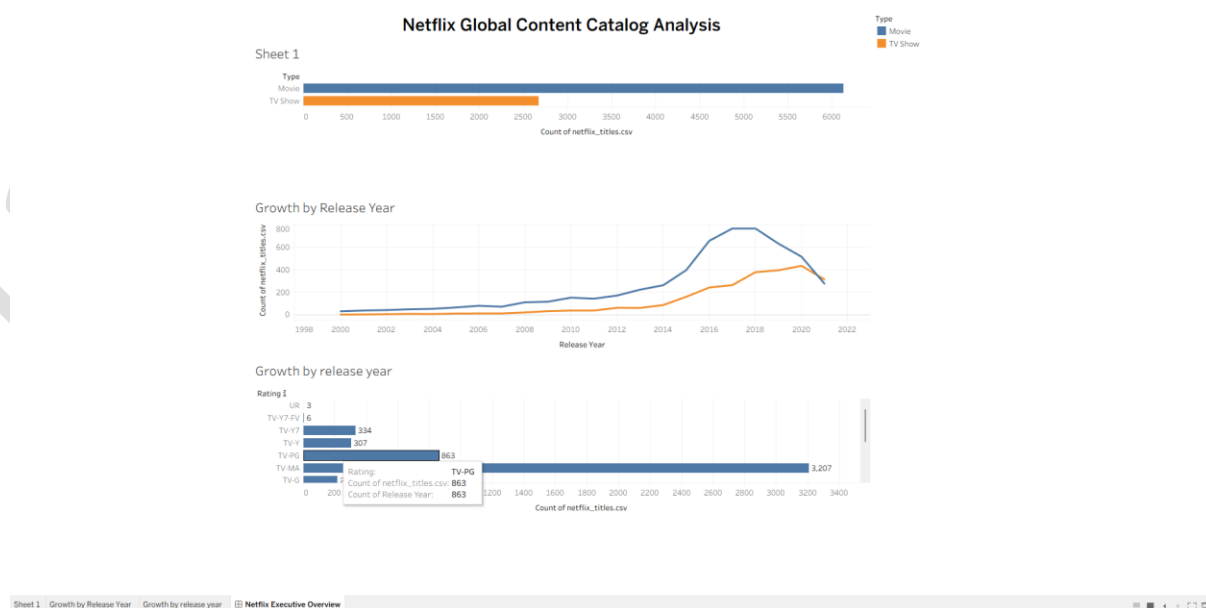
- Reset / Revert: Click the Revert arrow on the presentation toolbar to reset all filters back to their initial state



Step 2: Live Navigation and Interactivity During Presentation

While in Presentation Mode, test the following live controls:

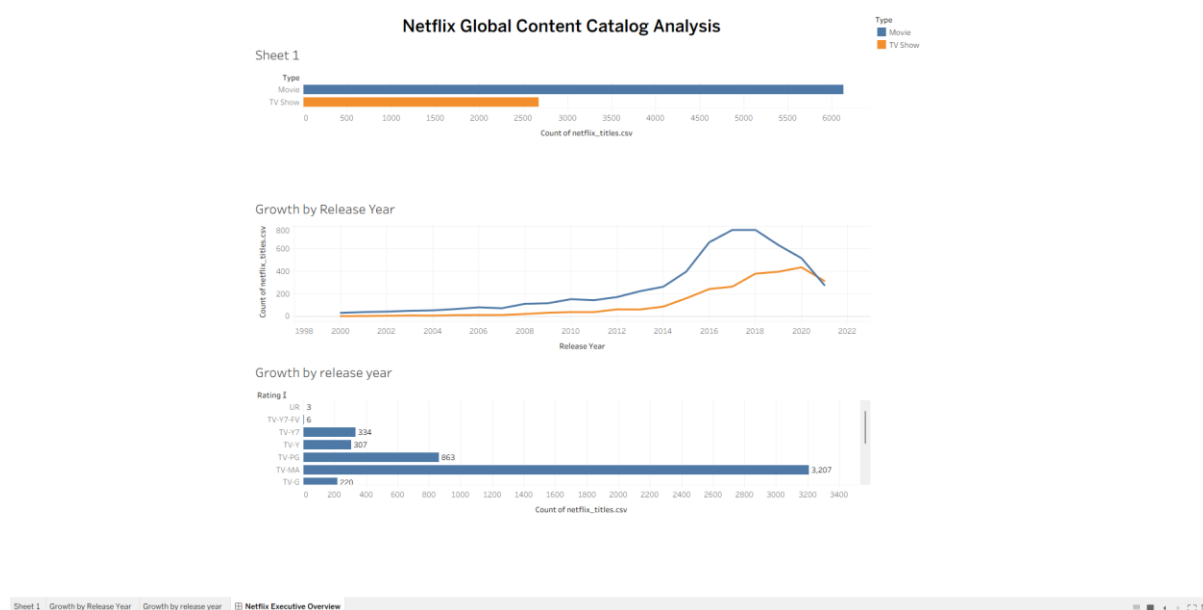
1. Hover Exploration:
 - o Move your mouse over individual data marks (e.g., specific bars in Content Type Split or peak data points in Growth by Release Year).
 - o Confirm that tooltips render cleanly with exact counts and dates.



2. Interactive Filtering:
 - o If quick filters or legends are placed on the dashboard, click to filter or highlight categories live.
3. Multi-Sheet Navigation:
 - o Move your mouse to the bottom-left corner to reveal the subtle navigation toolbar.
 - o Use the Next Sheet (↞) and Previous Sheet (↞) arrows (or the Page Up / Page Down keys) to cycle between individual worksheets and dashboards seamlessly.

Step 3: Exit Presentation Mode

- Press the Esc key on your keyboard (or click the Exit Presentation Mode icon in the bottom-left toolbar) to return to standard authoring mode.



C: Publish Dashboards to Tableau Public

Compatibility Note: You need to use "Tableau Desktop Public Edition to implement this task.

Local Environment Strategy: When completing this task in a local environment without active server credentials or cloud sync, the standard industry and academic deliverable for Tableau Public is a Tableau Packaged Workbook (.twbx) pre-configured with extracted data, metadata properties, and view presets ready for immediate web deployment or local grading.

Objective: Replace the cloud-dependent "Publish to Tableau Public" workflow by compiling your visualizations, calculations, formatting metadata, and data extract into a single, offline Tableau Packaged Workbook (.twbx)

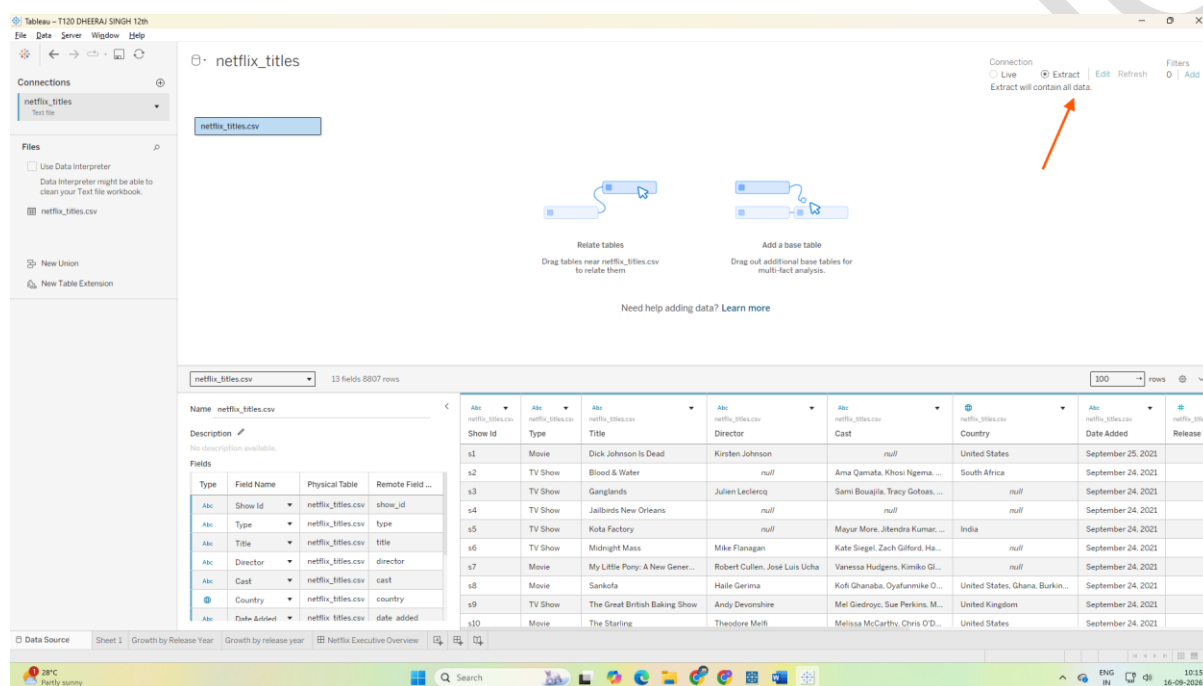
A standard Tableau workbook (.twb) contains only XML layout instructions and points to the external file path of netflix_titles.csv. If you move that CSV or share the .twb with an instructor or peer, the connection breaks. Generating a .twbx archives the underlying data extract directly inside the file container, enabling fully interactive, offline distribution without web accounts or server installations.

Step 1: Confirm Data Extract Status

A packaged workbook works best when the underlying data is extracted rather than referenced as a live text file.

1. Check the top-left Data Pane where netflix_titles is listed:
 - o If you see two linked cylinders (), the data is already extracted. Proceed to Step 2.
2. If you see a single cylinder ():
 - o Go to the Data Source tab in the bottom-left corner.
 - o In the top-right corner, select the Extract radio button.

o Click back onto your Netflix Executive Overview dashboard tab.

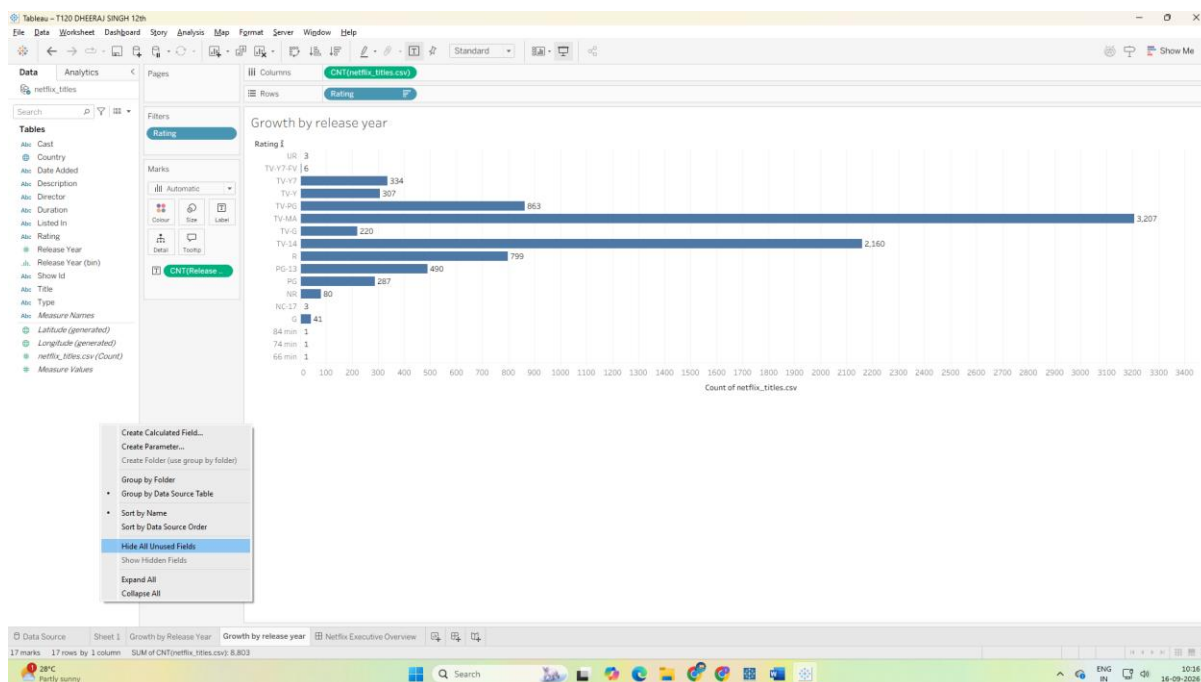


Step 2: Prune Unused Columns to Minimize File Footprint

To keep the packaged archive compact:

1. In the Data Pane, click the drop-down menu icon (small down-arrow next to the search bar).
2. Select Hide All Unused Fields.

o Outcome: Columns not used in your visualizations or calculations (e.g., full text descriptions or raw cast lists, if unused) will be excluded from the package, reducing disk space and load times.

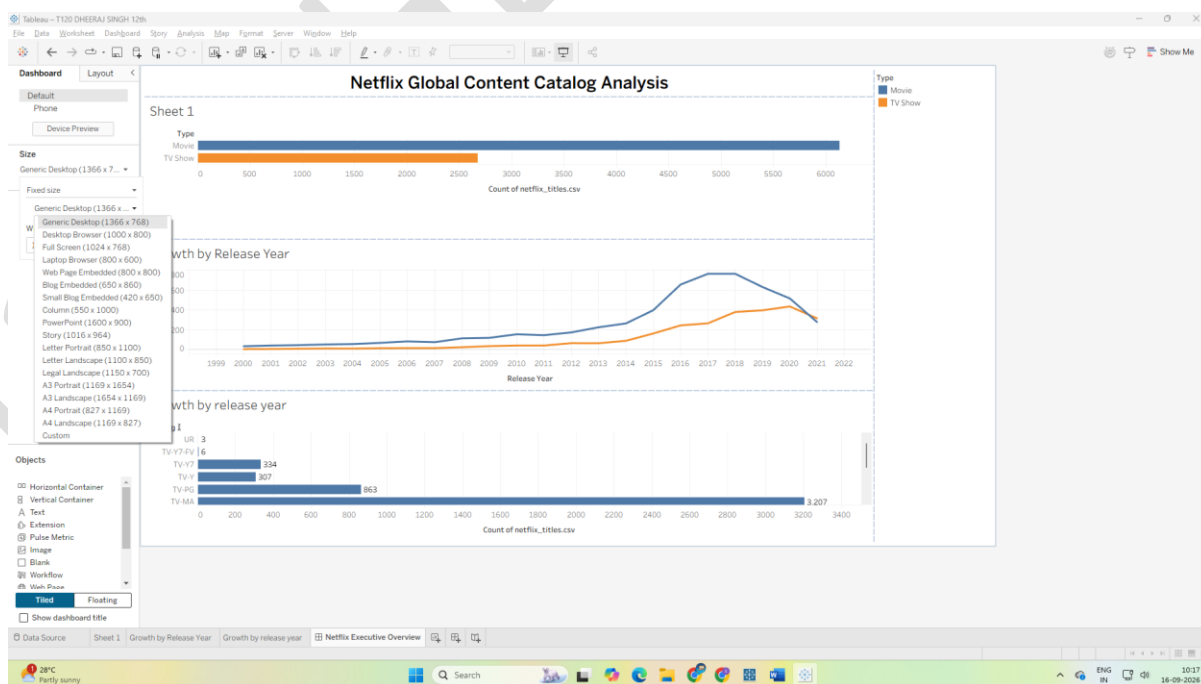


Step 3: Pin the Default Landing Tab

When someone opens your packaged file, Tableau opens whatever tab was active when the file was saved:

1. Click on the Netflix Executive Overview tab at the bottom so it is active.
2. In the left-hand Dashboard panel, ensure Size is set to Fixed Size (e.g., 1366 x 768 px or A4 Landscape).

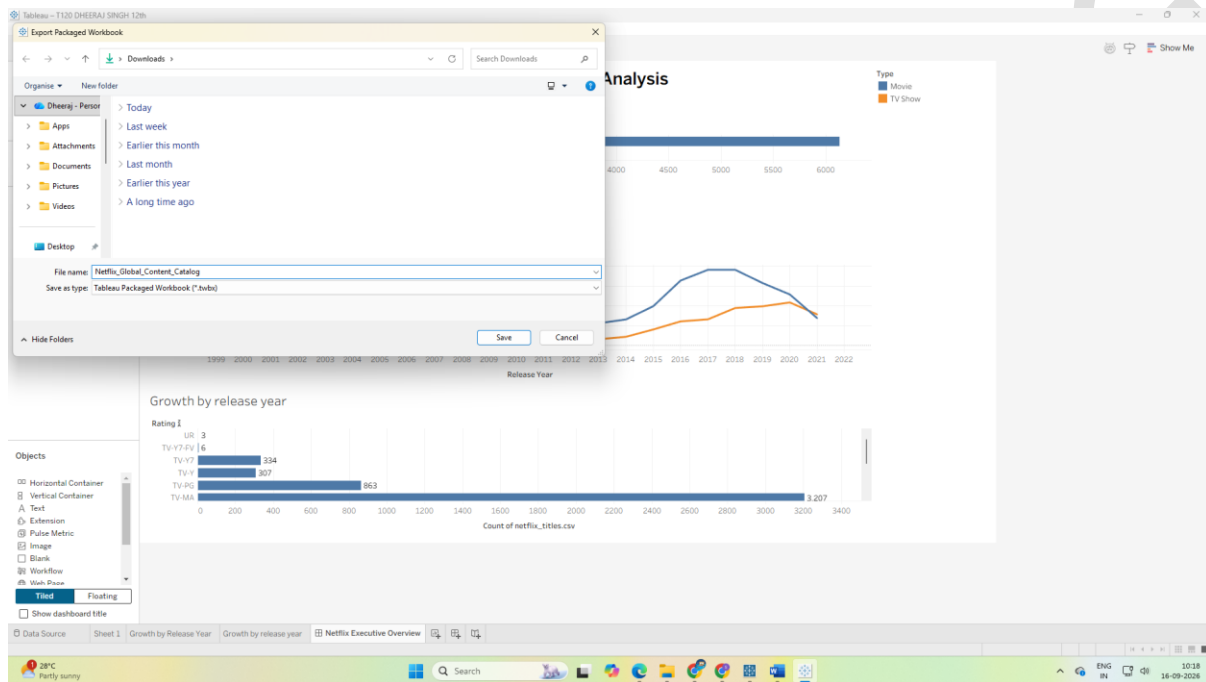
o Why: Fixed sizing prevents web browsers on different screen resolutions from distorting the chart layout or causing unwanted scrollbars.



Step 4: Export the Standalone Packaged Workbook (.twbx)

1. Go to the top menu bar and click File --> select Export Packaged Workbook... (or File --> Save As...).
2. In the file save window: o File name: Netflix_Global_Content_Catalog.twbx o Save as type: Select Tableau Packaged Workbook (*.twbx)
3. Click Save.

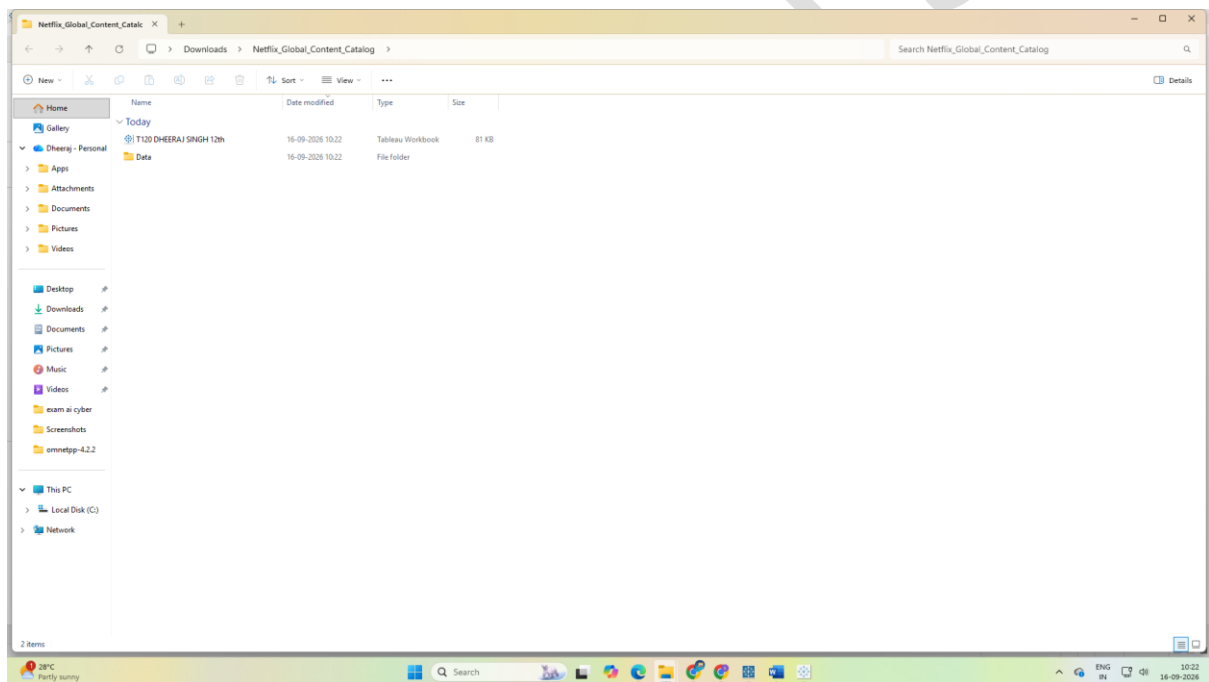
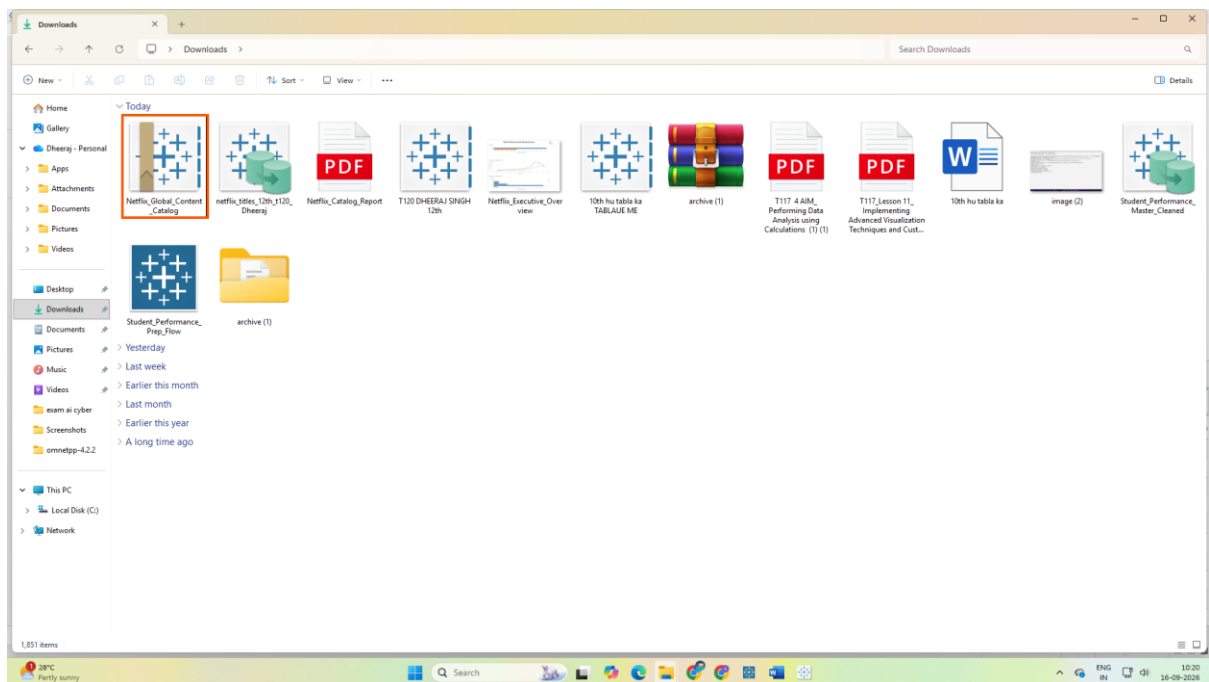
What this file contains: Both the .hyper data extract and the complete visual design (XML code, filters, formatting), making it completely independent of your local netflix_titles.csv file.

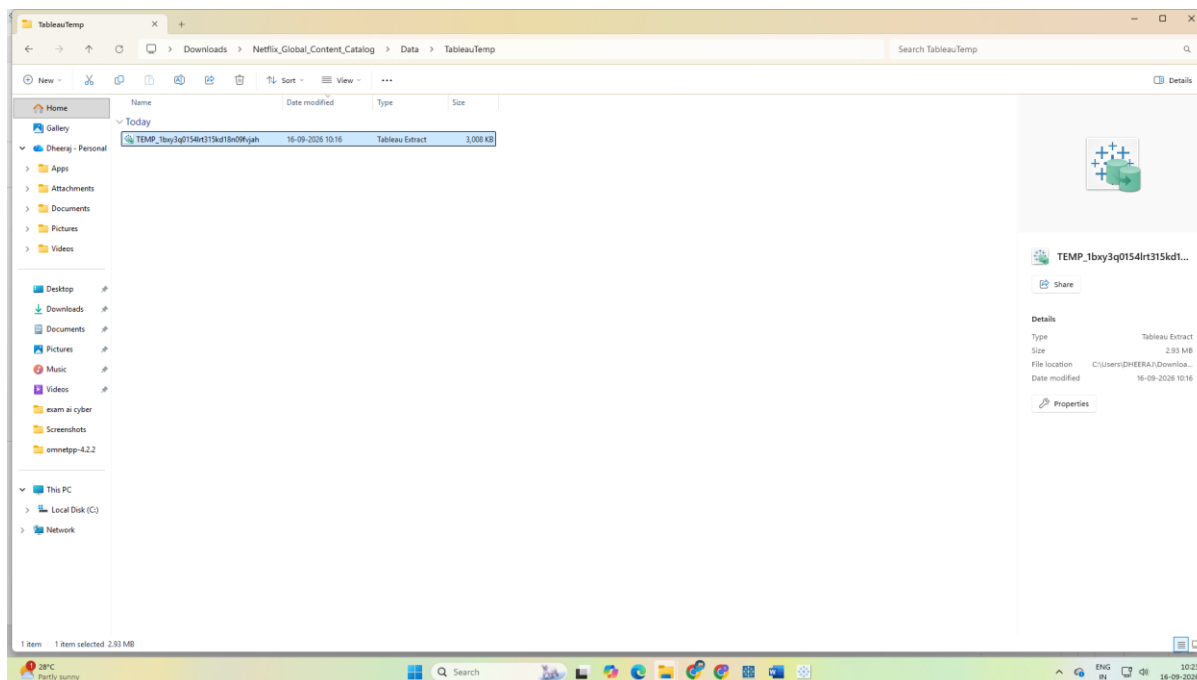


Step 5: Verify Archive Integrity Locally

To confirm that the package is completely self-contained:

1. In your operating system file explorer (Windows Explorer), locate Netflix_Global_Content_Catalog.twbx.
2. Notice the file icon: a standard Tableau workbook icon with a small orange package emblem.
3. (Technical Verification) A .twbx file is internally a standard ZIP archive. If you rename the extension from .twbx to .zip and open it, you will find:
 - o The .twb (XML schema and visualization code).
 - o A Data folder containing the compressed .hyper extract.

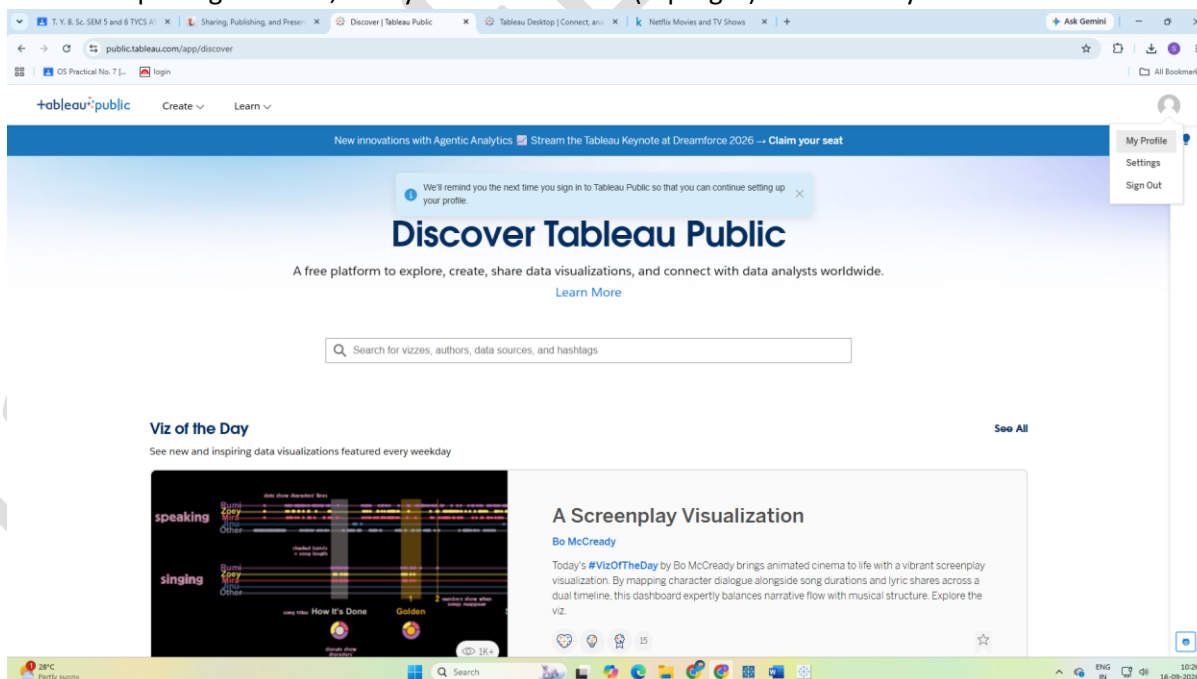




4. This confirms the workbook can be moved to any computer or shared via USB/email without needing netflix_titles.csv.

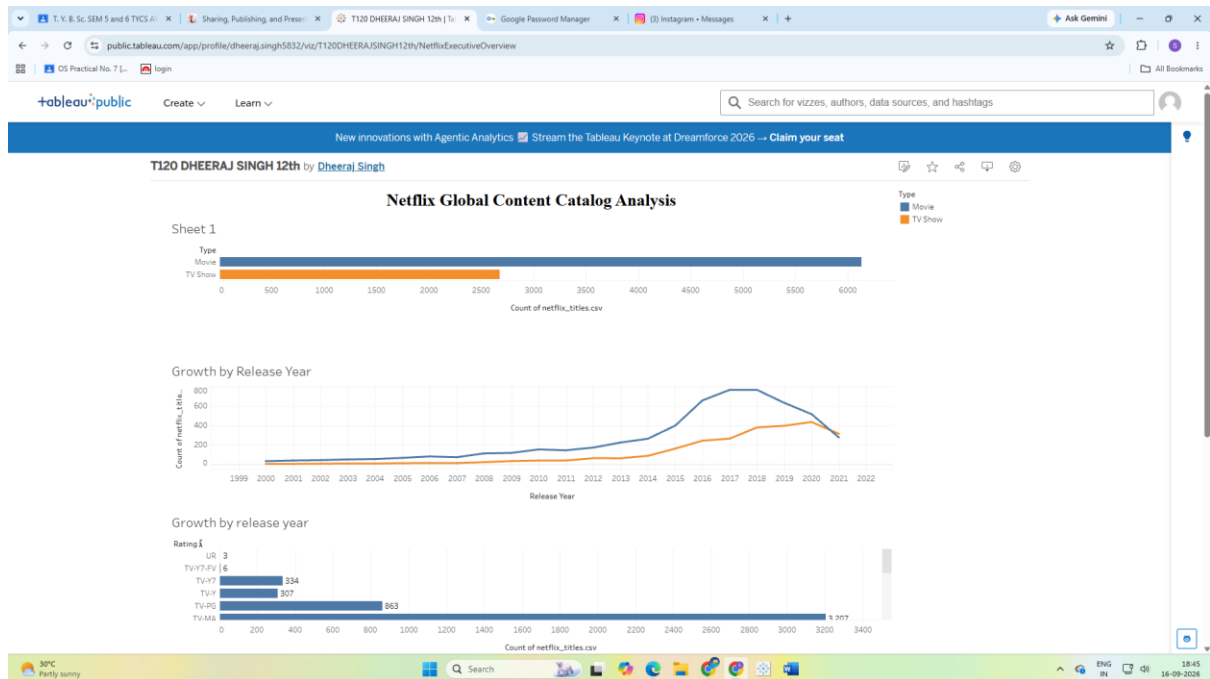
Additional Step : Publish to Tableau Public via Web Interface[if you have tableau public login]

1. Open your web browser and go to public.tableau.com.
2. Sign in to your free Tableau Public account (or click Sign Up to create one).
3. In the top navigation bar, click your Profile Avatar (top-right) --> select My Profile.



4. Click the Create / Upload button (or the Upload a Viz option).
5. Drag and drop your saved Netflix_Global_Content_Catalog.twbx file into the upload area.

6. Once the upload finishes, your interactive dashboard is live on Tableau Public. 7. On the published web page:



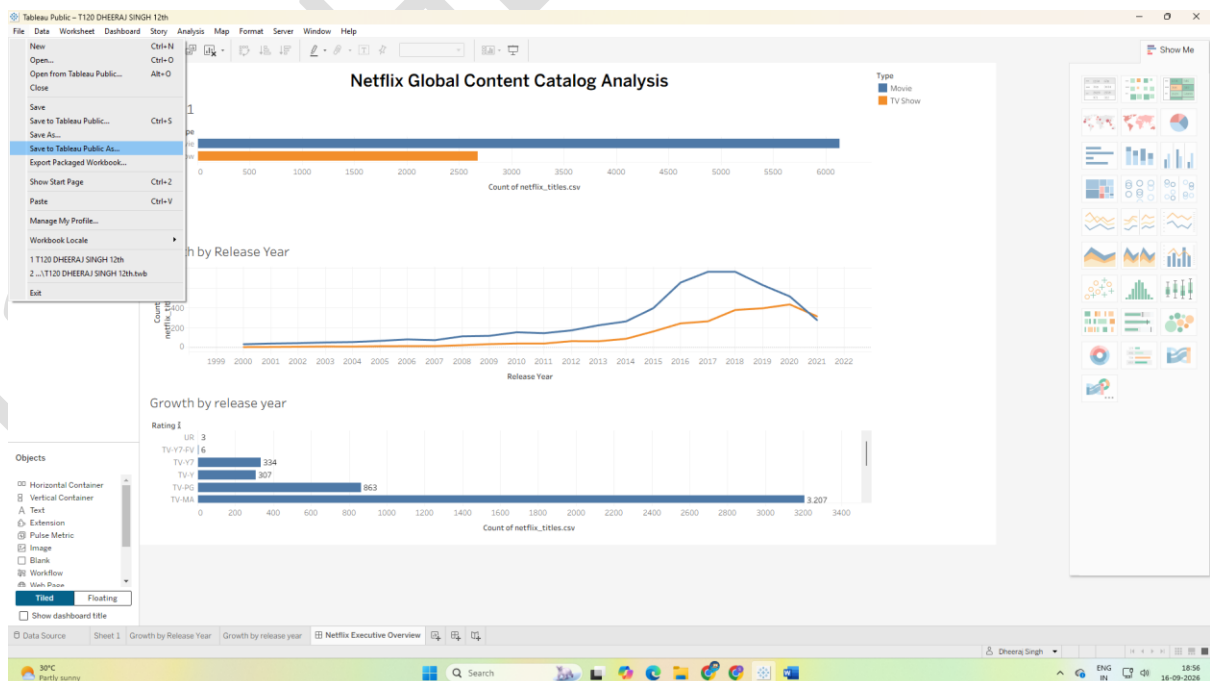
- Test Interactivity: Click on the Movie or TV Show bars to confirm web crossfiltering works.

- Retrieve Links: Click the Share button below the dashboard to copy:

o Direct URL: For resumes, LinkedIn, or portfolio submissions. o LINK:

https://public.tableau.com/views/T120DHEERAJ SINGH12th/NetflixExecutiveOverview?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

o Embed Code: For embedding into HTML websites or blogs.



D: Share Dashboards via Tableau Server / Tableau Cloud

Objective: Implement the local offline equivalent of enterprise sharing by configuring sheet visibility, distribution governance, and simulating end-user consumption using free distribution tools like Tableau Reader.

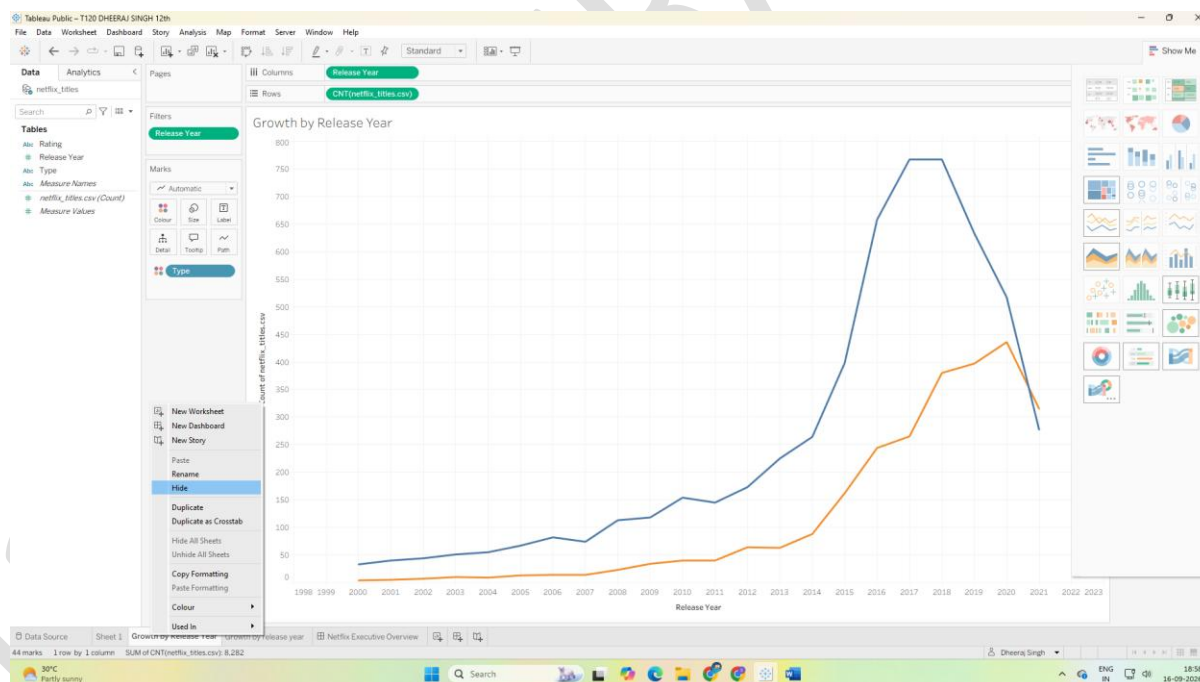
In corporate settings without centralized Tableau Server or Cloud infrastructure, workbooks must still be shared securely and cleanly. This requires preventing end-users from seeing messy draft sheets, securing raw underlying data from casual alteration, and verifying that the deliverable functions properly for users who do not own an authoring license.

Step 1: Clean Up Workbook Tabs (Sheet Governance)

In enterprise deployments, raw working sheets should never be visible to business stakeholders. Only finalized executive dashboards should appear in the navigation bar.

1. Hide each underlying worksheet:

- Right-click the Content Type Split tab --> select Hide Sheet.
- Right-click Growth by Release Year --> select Hide Sheet.
- Right-click Top 10 Ratings --> select Hide Sheet.
- (Note: If "Hide Sheet" is grayed out, ensure the sheet is currently placed inside your Netflix Executive Overview dashboard).



- Verify the Layout: Look at the bottom tab bar. Only the Netflix Executive Overview dashboard tab should now be visible.

Step 2: Configure Navigation and Sheet Tabs Settings

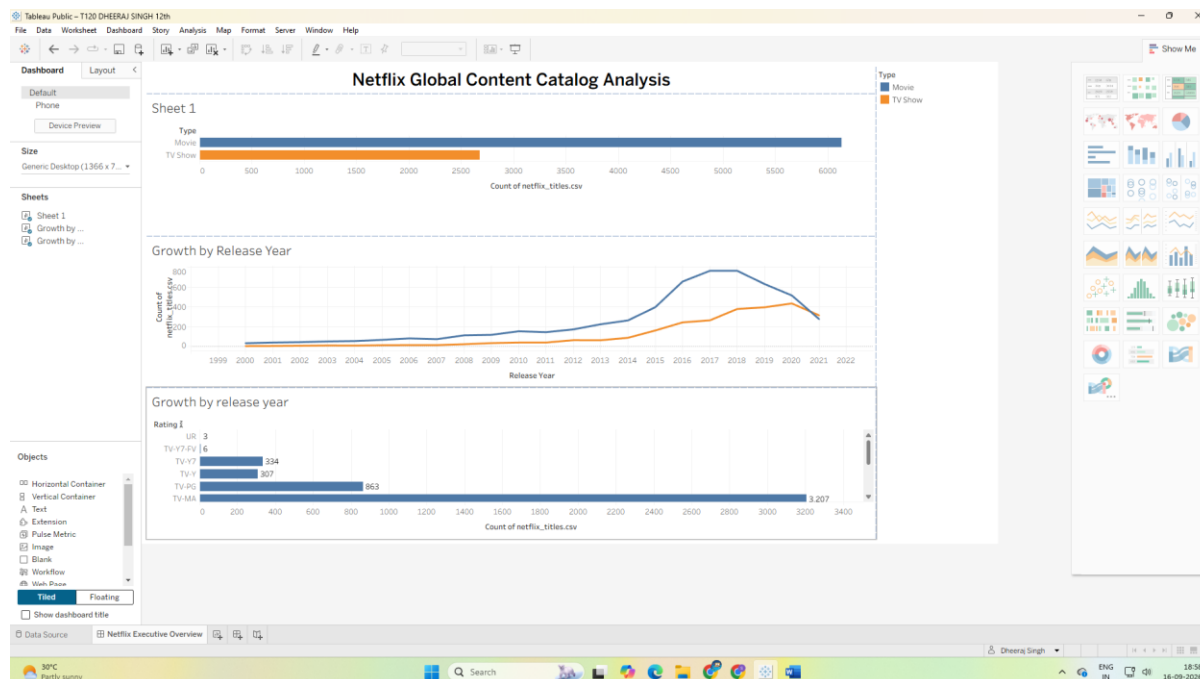
When stakeholders open the file, configure how Tableau displays navigation:

1. Go to the top application menu bar and click Window.

2. Look at the Show Sheet Tabs toggle:

o If you have only one executive dashboard, you can leave it unchecked or keep it clean so the dashboard spans the entire interface.

3. Verify that the window controls are clean and unobstructed by unused toolbars.



Step 3: Understand the Tableau Reader Distribution Mechanism

When Tableau Server or Tableau Cloud is unavailable, organizations distribute dashboards via Tableau Reader:

• What is Tableau Reader?

- o A free, standalone desktop viewer application for Windows and macOS.
- o Downloadable at no cost by any stakeholder or client without a license key.

• Consumer Capabilities in Tableau Reader:

- o Interactivity: Viewers can click marks, filter categories, hover for tooltips, and change parameters.
- o Zero Accidental Edits: Viewers cannot move charts, alter formulas, break calculations, or modify visual aesthetics.
- o File Requirement: Tableau Reader opens only .twbx (packaged) workbooks with extracted data, ensuring stakeholders never experience "data source not found" errors.

Step 4: Simulate Consumption via Tableau Reader / Fresh Session

To verify that your workbook functions as a true enterprise-ready deliverable:

1. In Tableau Desktop, press Ctrl + S (or go to File > Save) to save your packaged workbook (Netflix_Global_Content_Catalog.twbx).
2. Close your current Tableau Desktop window.

3. Double-click the saved Netflix_Global_Content_Catalog.twbx file in your local file explorer to reopen it.

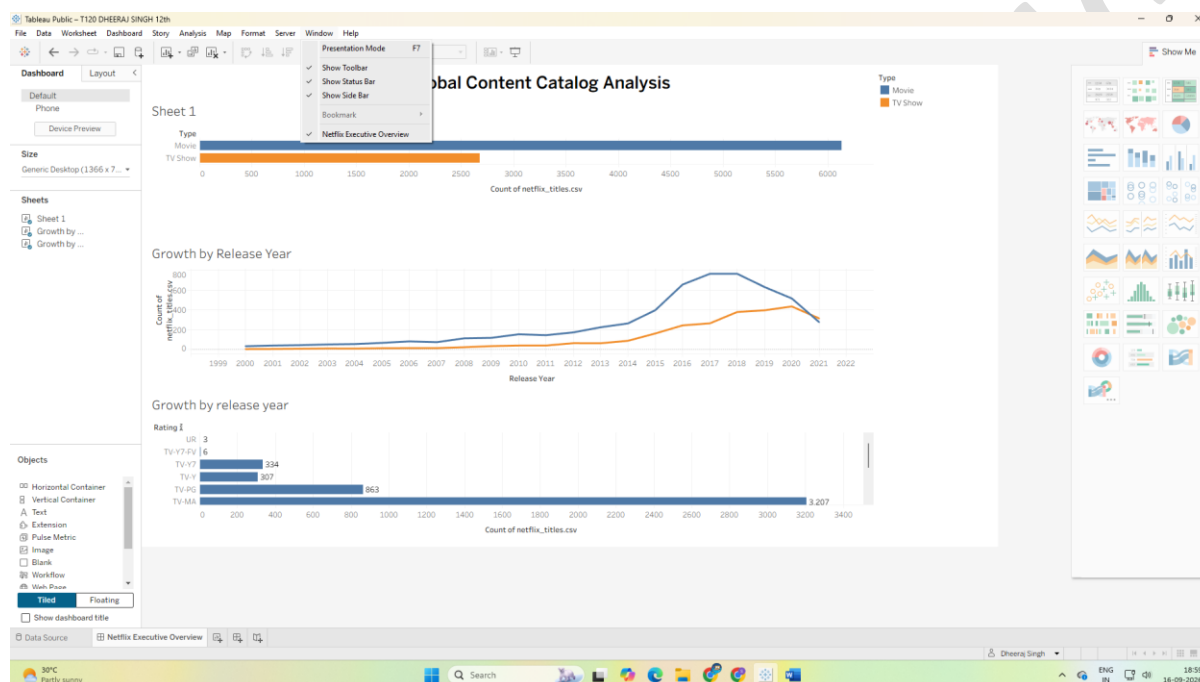
4. Audit the File:

o Does it open directly to the Netflix Executive Overview?

o Are the underlying calculation/raw sheets hidden from view?

o Does it open instantly without asking for database passwords or CSV file paths?

5. If yes to all, your deliverable is structured correctly for offline distribution.



E: Enable User Interaction on Shared Dashboards

Compatibility Note (Tableau Desktop 2026.2): Fully supported natively in all editions (Free, Trial, and Professional).

Objective: Configure interactive features including Dashboard Filter Actions, Highlight Actions, Viz in Tooltips, and URL Web Actions on the Netflix dashboard, so that offline stakeholders can explore the data dynamically without needing Tableau Server or authoring tools.

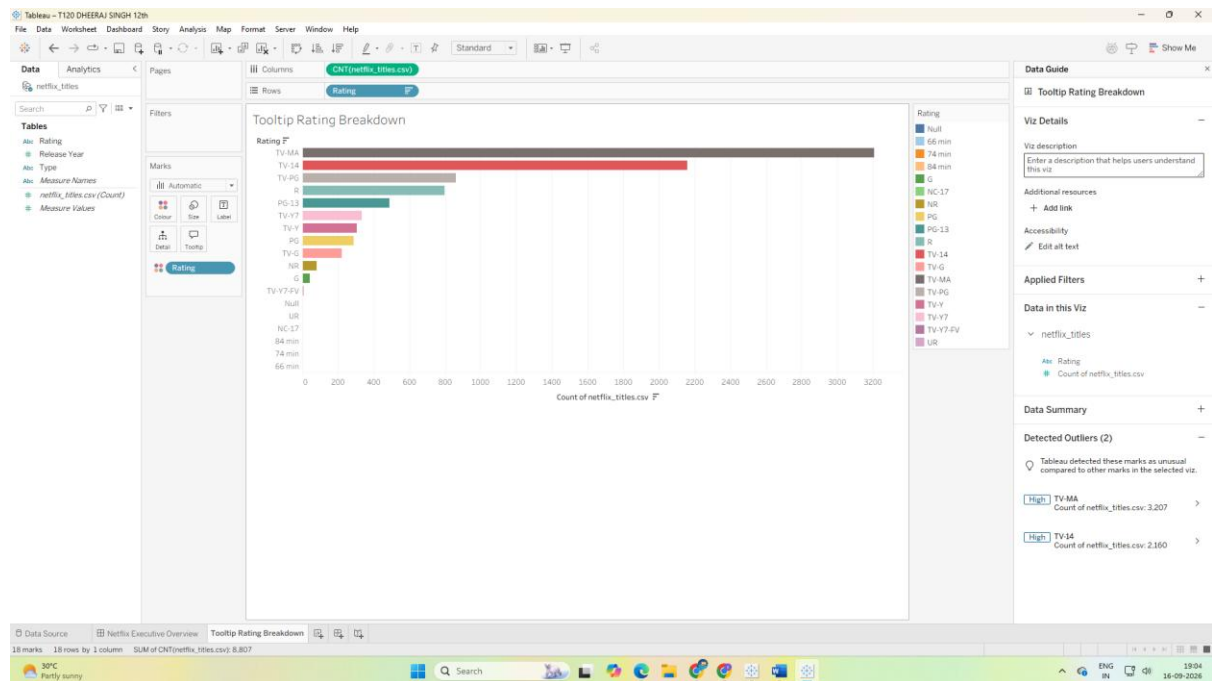
An executive dashboard should not be a static grid of charts. Adding interactive actions allows users to click a category to filter time-series trends across other charts, hover over a summary bar to see a detailed micro-chart embedded in the tooltip, and launch contextual external searches directly from data marks.

Step 1: Create the Detail Sheet for Viz in Tooltip

A Viz in Tooltip dynamically renders an embedded secondary chart inside the hover tooltip of a primary chart.

1. Create a new worksheet named: Tooltip Rating Breakdown.
2. Drag Rating onto the Rows shelf.

3. Drag netflix_titles.csv (Count) onto the Columns shelf. 4. Drag Rating onto the Color tile on the Marks



Step 2: Embed the Viz into the Main Tooltip

1. Return to your Content Type Split worksheet.

2. On the Marks card, click the Tooltip tile.

3. In the Edit Tooltip dialog:

o Delete default text if desired, then click Insert in the top-right corner of the editor.

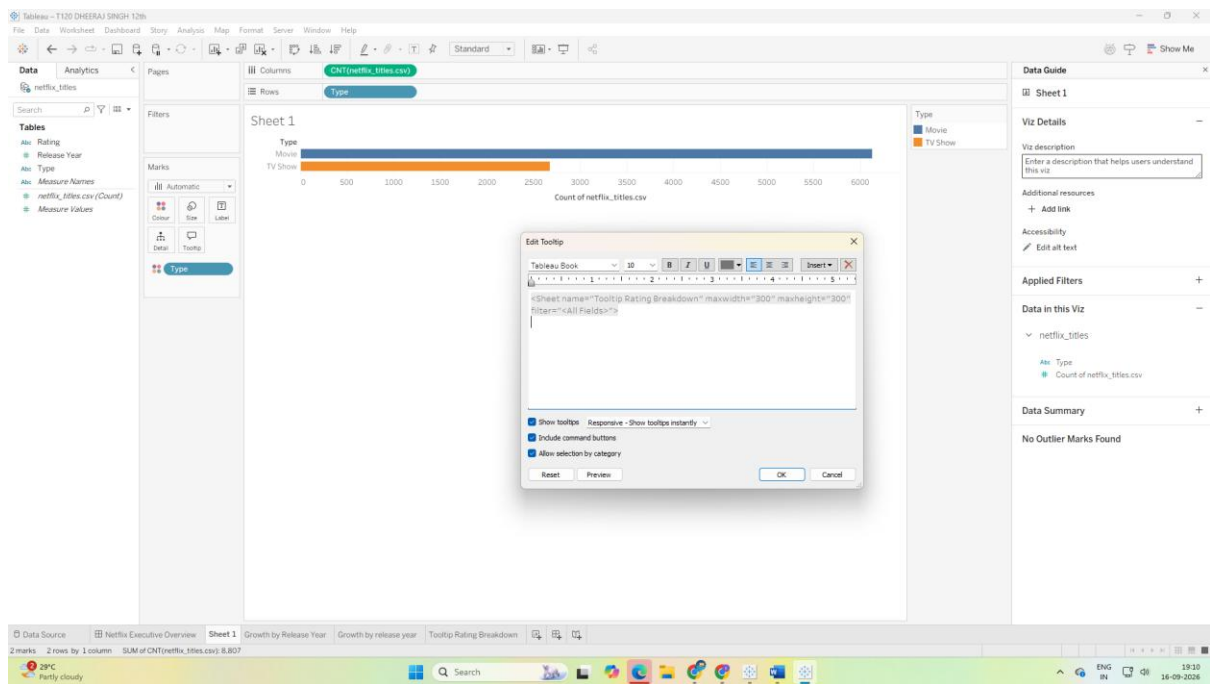
o Select Sheets --> Tooltip Rating Breakdown.

o Tableau inserts the code tag:

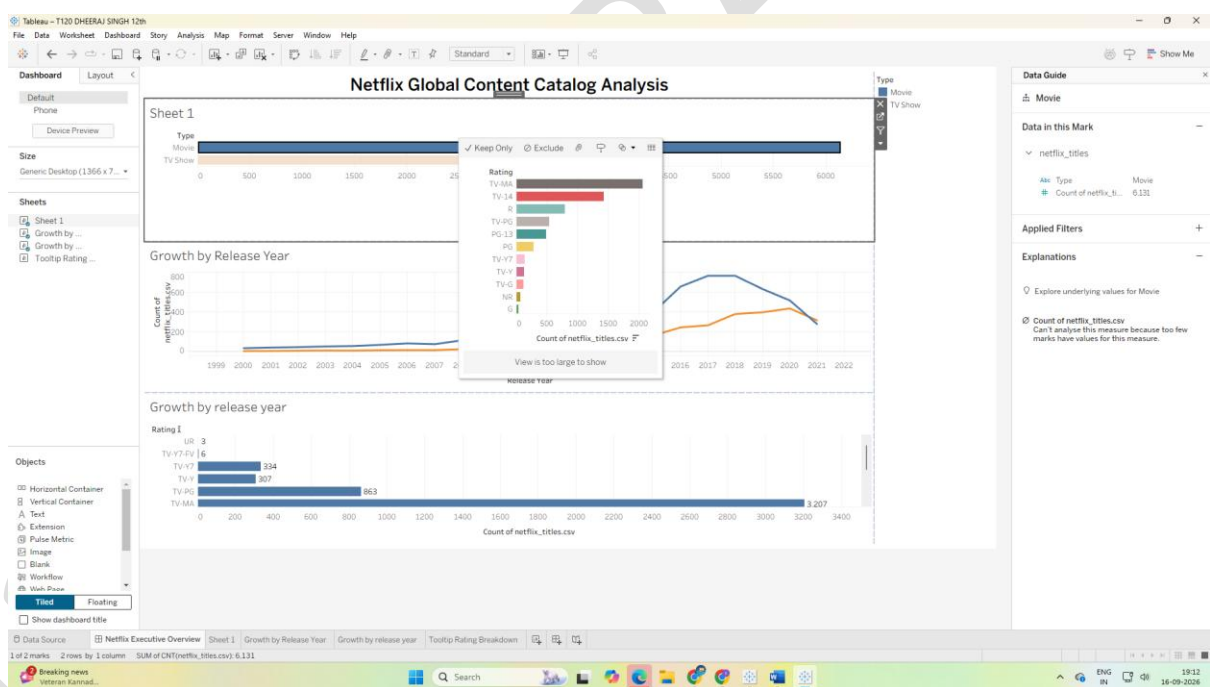
<Sheet name="Tooltip Rating Breakdown" maxwidth="300"

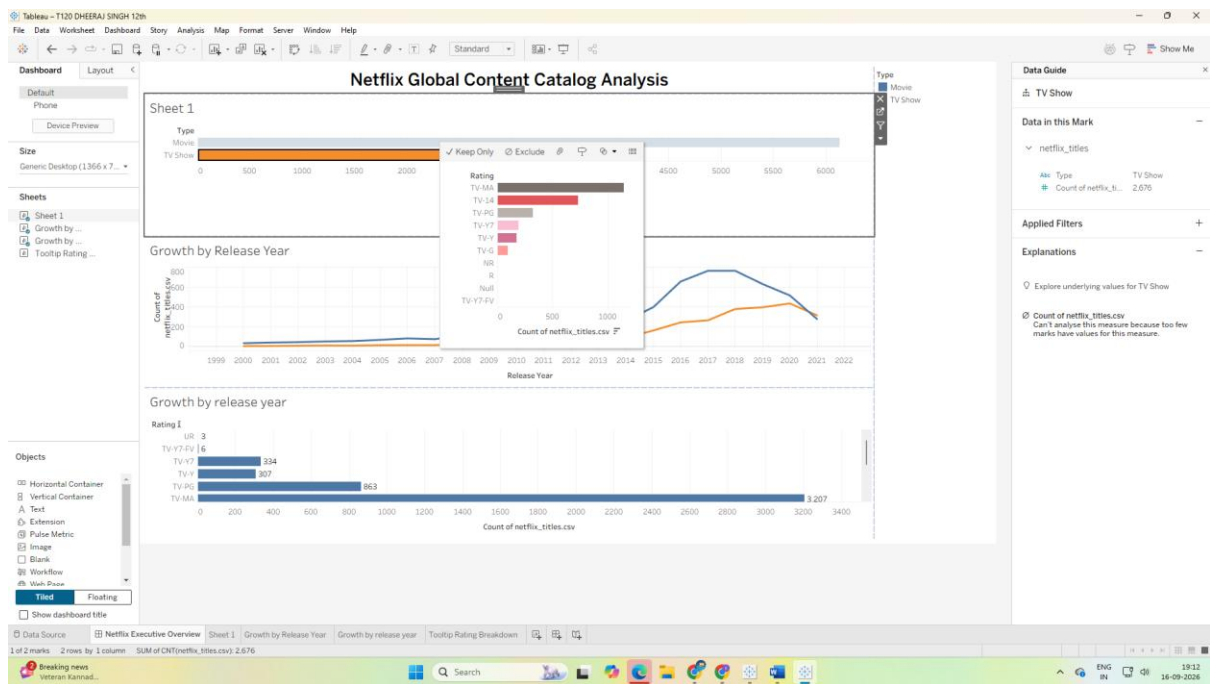
maxheight="300" filter="<All Fields>">

4. Click OK.

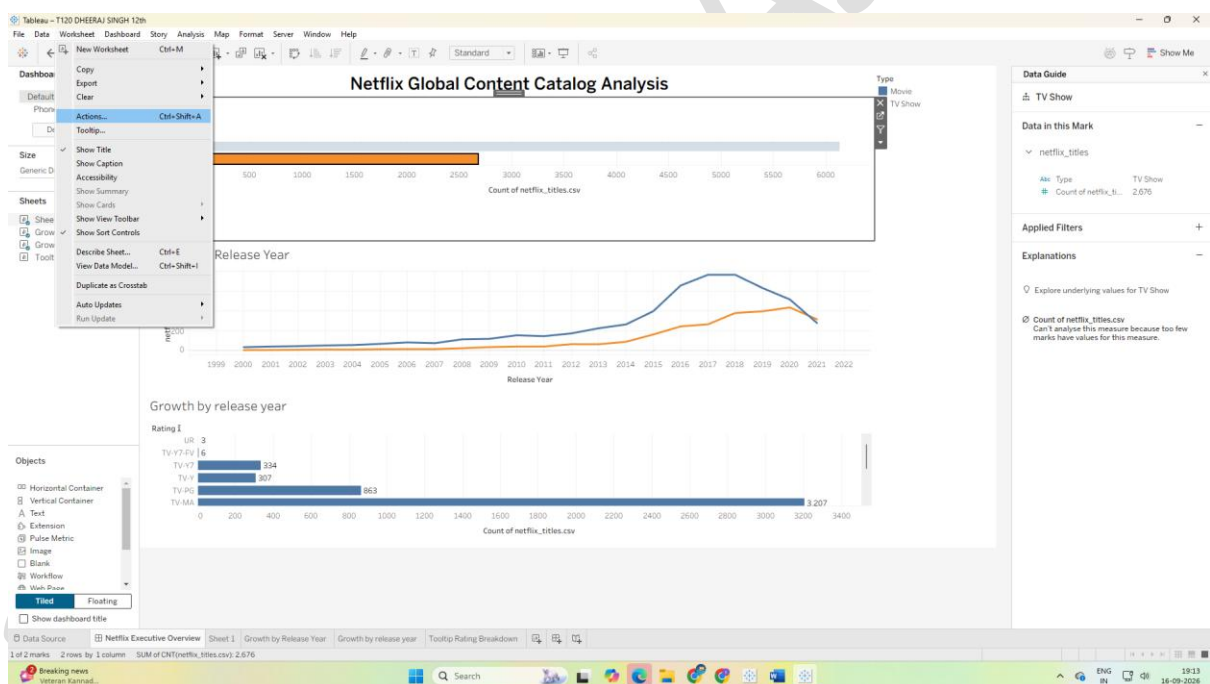


4. Switch back to your Netflix Executive Overview dashboard and test it:
 - Hover your mouse over the Movie bar: you will see a mini-bar chart inside the tooltip showing the rating distribution for movies only.





- Hover over the TV Show bar: the mini-bar chart instantly changes to show TV show ratings (TV-MA, TV-14, etc.).

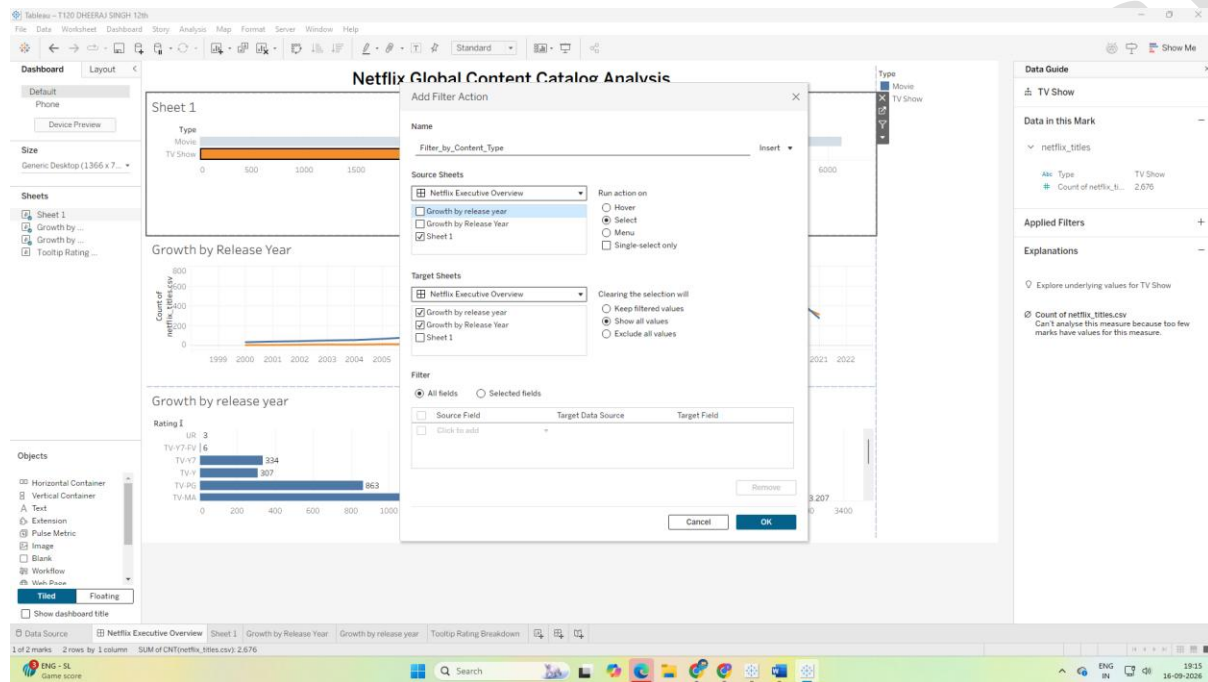


Step 3: Configure Dashboard Filter Actions (Cross-Filtering)

Clicking a mark in one chart will filter the other charts on the dashboard.

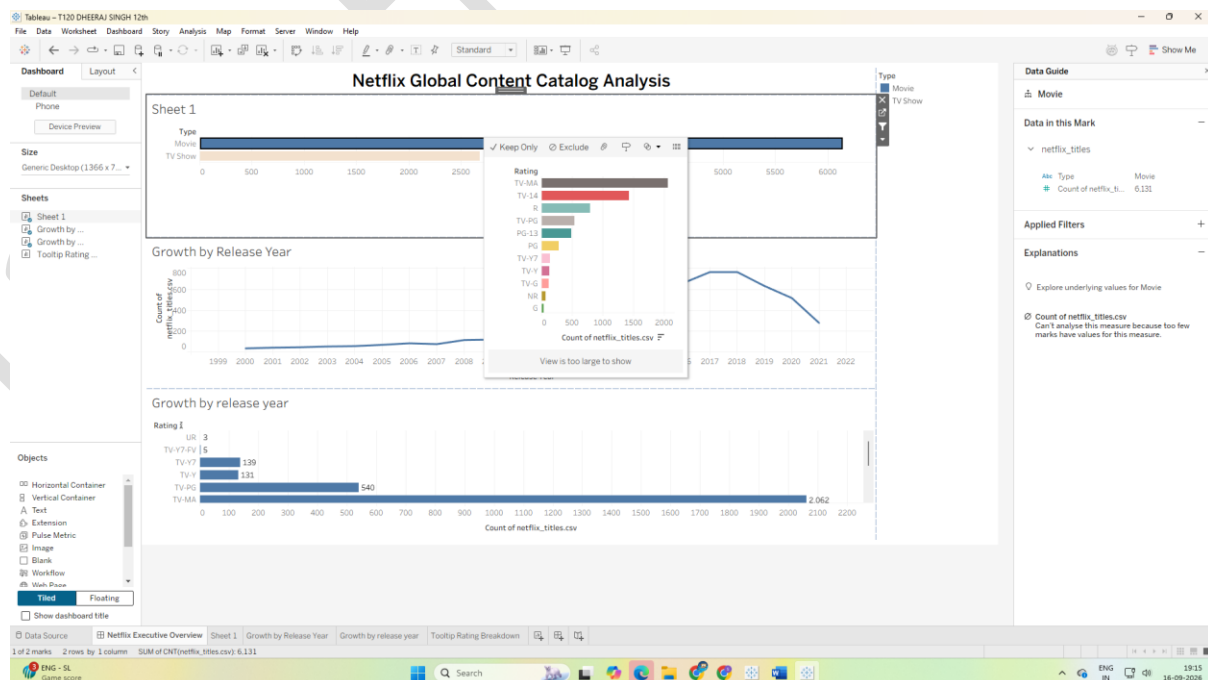
1. Open your Netflix Executive Overview dashboard.
2. From the top menu bar, select Dashboard --> Actions... (or press Ctrl + Shift + A).
3. In the Actions dialog, click Add Action --> select Filter....
4. Configure the Filter Action:

- o Name: Filter_by_Content_Type
- o Source Sheets: Check Content Type Split. (Sheet 1)
- o Run action on: Select Select (triggers on click).
- o Target Sheets: Check Growth by Release Year and Top 10 Ratings.
- o Clearing the selection will: Select Show all values.
- o Target Filters: Leave as All Fields.



5. Click OK, then click OK again.

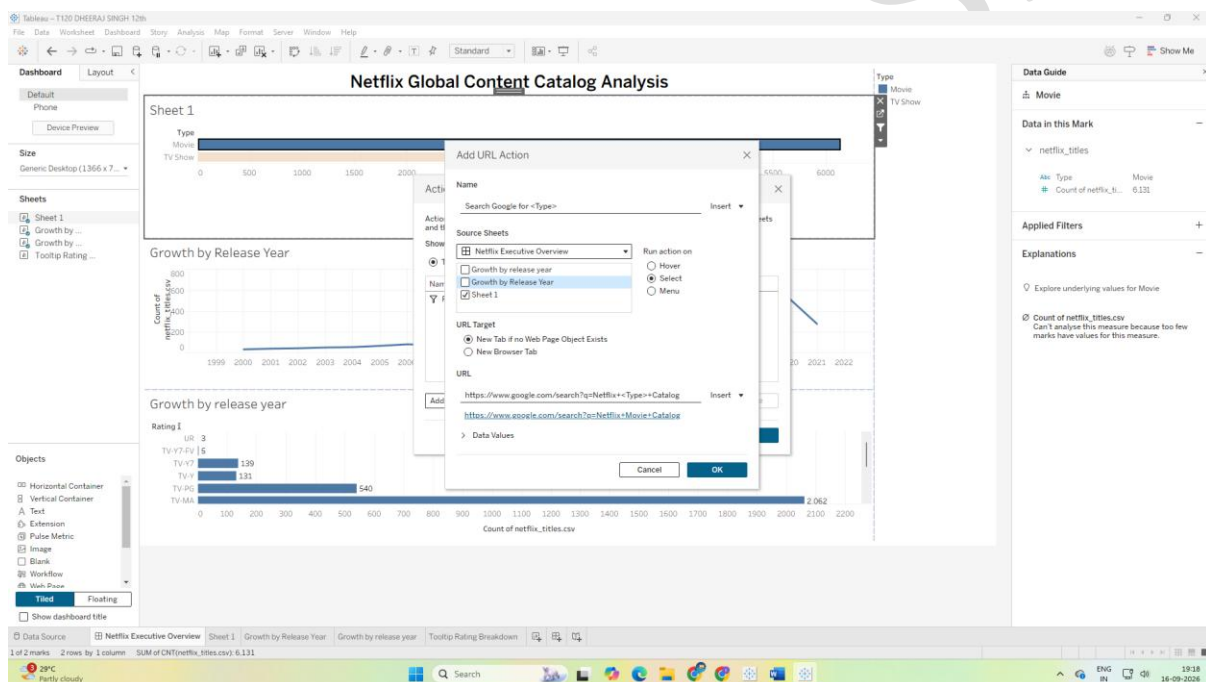
6. Test the Filter: Click the Movie bar; the line chart instantly updates to show release year trends exclusively for movies. Click it again to clear the filter.



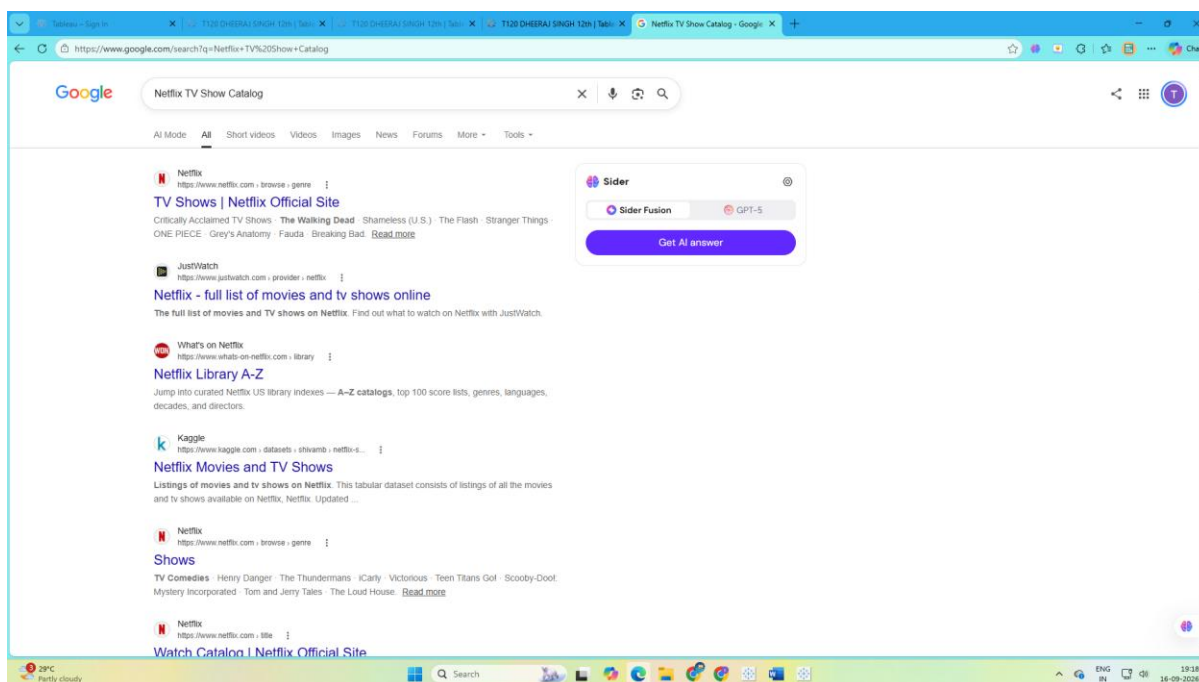
Step 4: Add a Contextual Web URL Action

URL Actions allow users to click a mark and open an external web page (e.g., searching Google or IMDb for a selected title or genre).

1. Go to Dashboard --> Actions... --> click Add Action --> select Go to URL....
2. Configure the URL Action:
 - o Name: Search Google for
 - o Source Sheets: Check Content Type Split.
 - o Run action on: Select Menu (adds a clickable link inside the tooltip).
 - o URL: Enter: <https://www.google.com/search?q=Netflix++Catalog>
 - o (Click the small arrow next to URL to insert the dimension dynamically).
3. Click OK, then click OK to close.



4. Hover over any bar and click the hyperlink menu item to verify that the browser opens the dynamic search query.



F: Manage Permissions and Access Control

Version & Environment Note: In standalone, unactivated trial, or local Tableau Desktop environments without an active connection to Tableau Server/Cloud, built-in user directories and the bottom-right "Filter as User" menu are disabled. To fulfill the core objective of RowLevel Security (RLS) and Role-Based Access Control (RBAC) locally, we simulate user personas and security clearance using dynamic parameters and conditional logic.

Objective: Build and test a local Row-Level Security (RLS) filter that restricts dataset visibility based on user roles and regional clearance.

In enterprise reporting, sensitive data must be filtered based on who is viewing the report (e.g., a regional analyst should only see local market data, while executive leadership sees the complete global catalog). Implementing security logic at the calculation and data source layer ensures single-workbook scalability without unauthorized data exposure.

Step 1: Create the User Persona Simulation Parameter

1. In the Data Pane on the left, click the small dropdown arrow next to the search bar (or right-click an empty space) --> select Create Parameter....

2. Configure the parameter properties:

- o Name: Simulated_User_Role

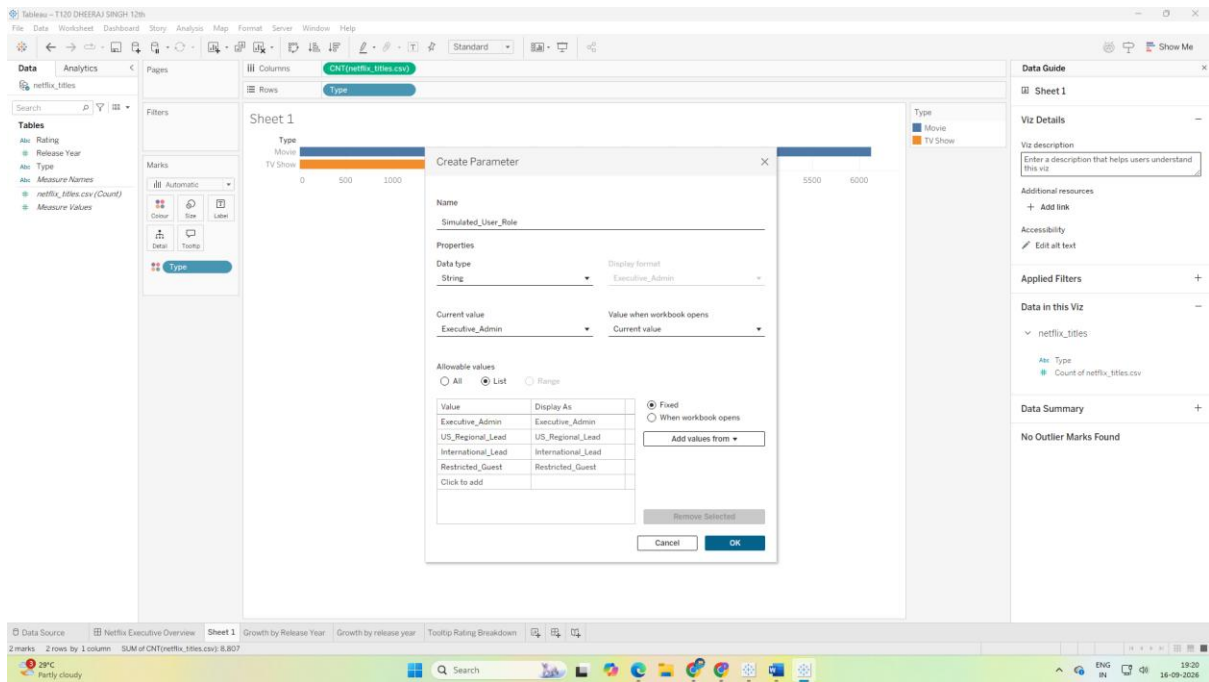
- o Data type: String

- o Allowable values: Choose List

- o Add four simulated corporate identities:

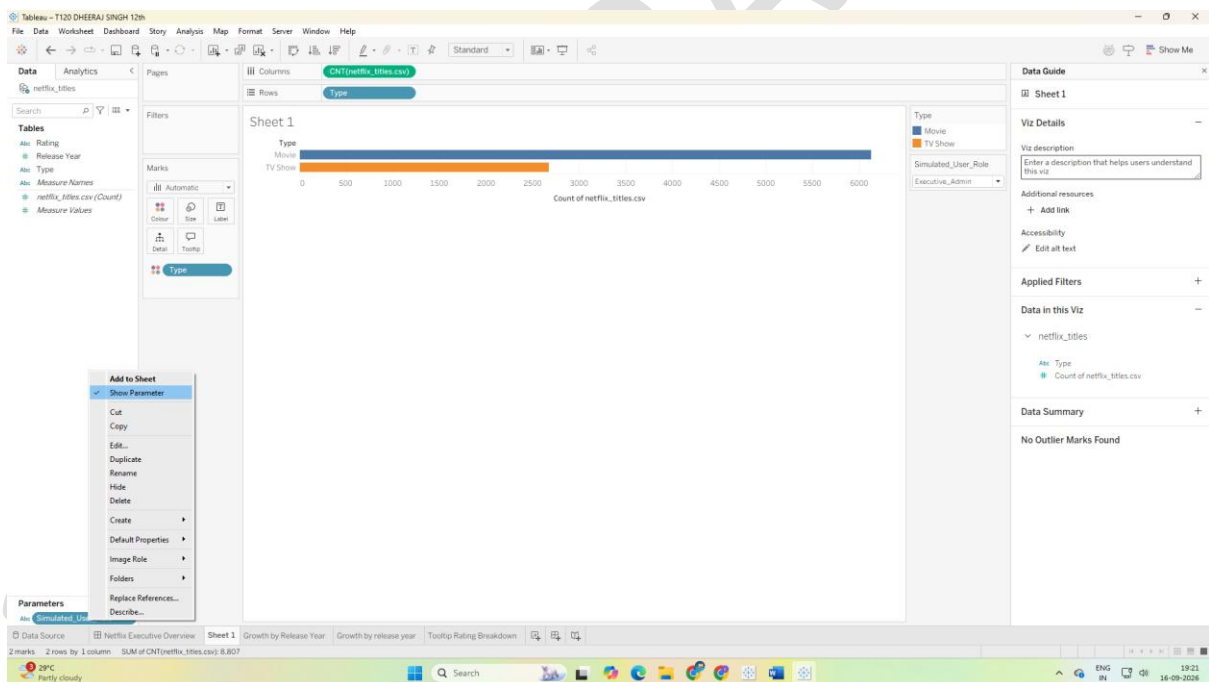
1. Executive_Admin 2. US_Regional_Lead 3. International_Lead 4. Restricted_Guest

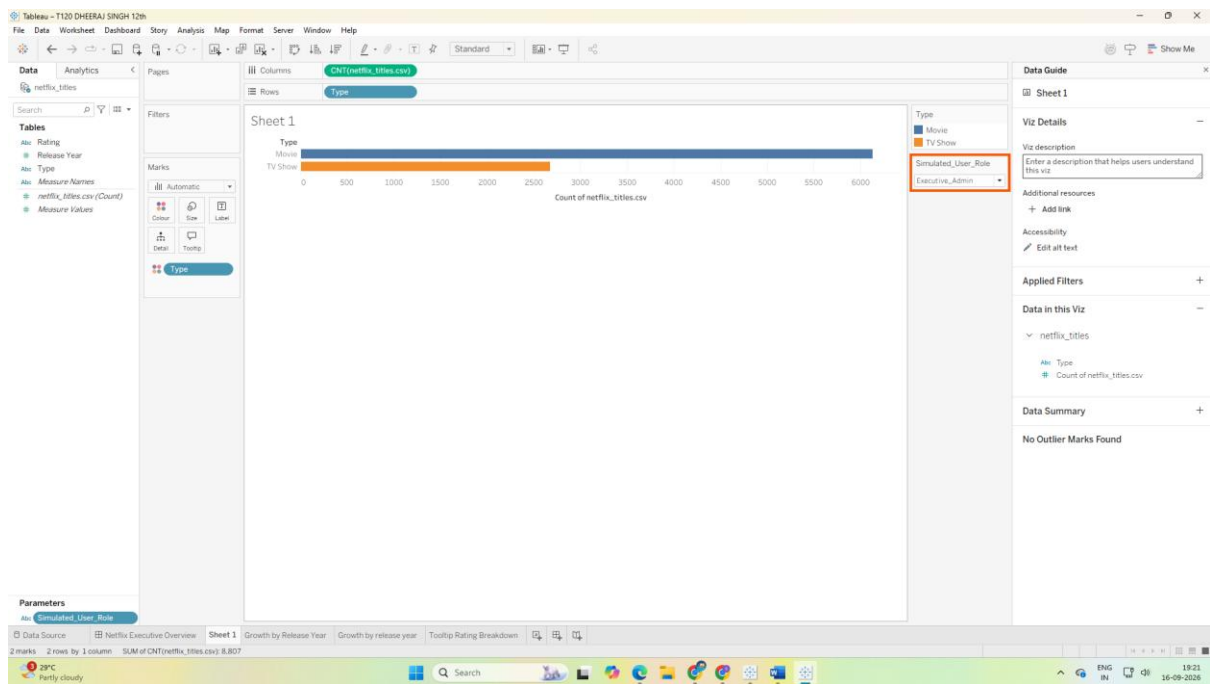
- o Current value: Set to Executive_Admin



3. Click OK.

4. In the Data Pane under Parameters, right-click Simulated_User_Role --> select Show Parameter.





Step 2: Write the Access Control Calculated Field

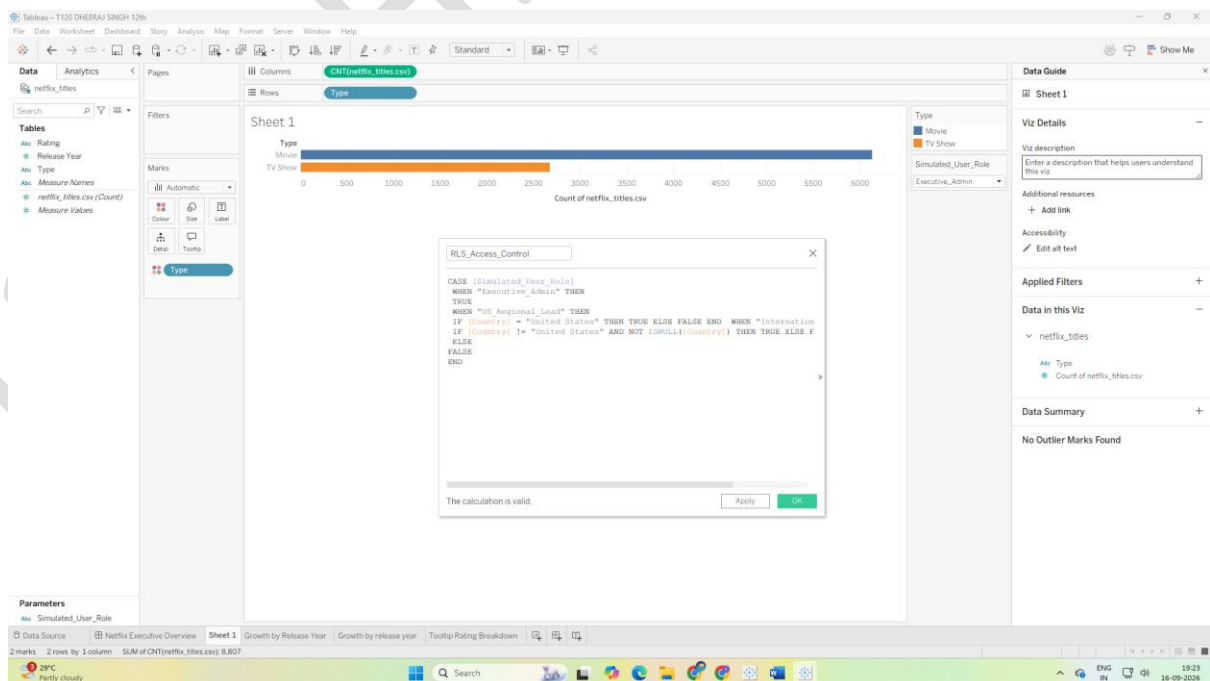
1. Right-click an empty space in the Data Pane --> select Create Calculated Field....

2. Configure the field:

o Field Name: RLS_Access_Control

o Tableau Formula:

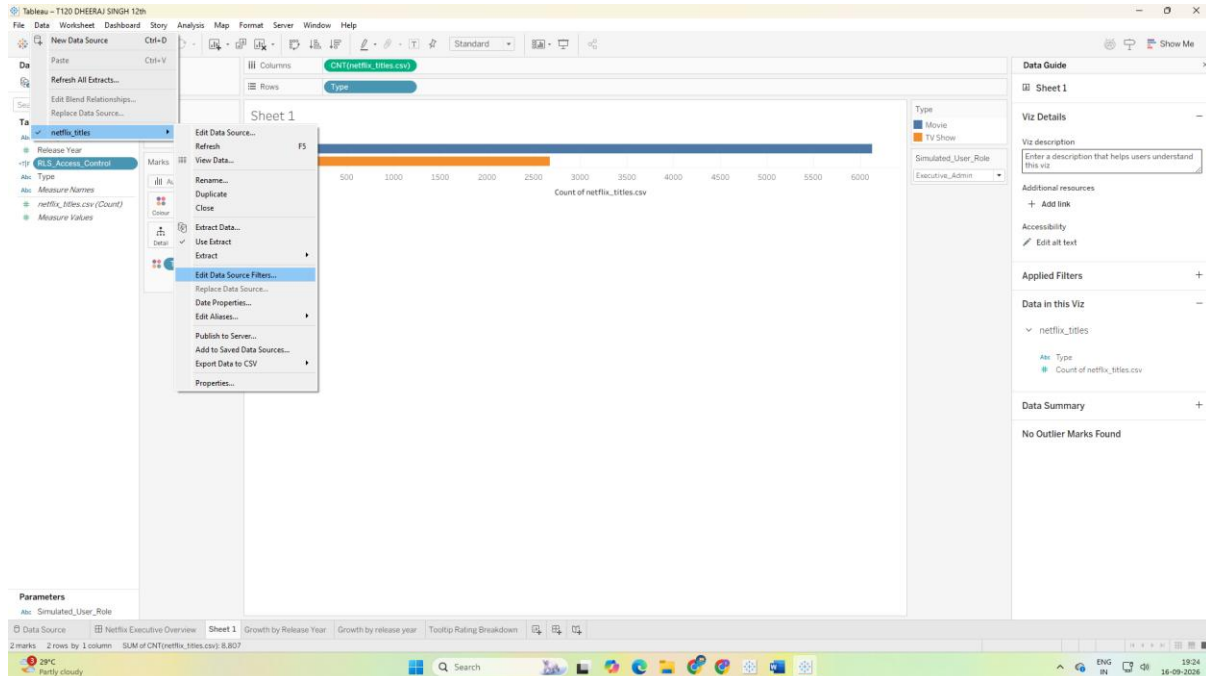
CASE [Simulated_User_Role] WHEN "Executive_Admin" THEN TRUE WHEN "US_Regional_Lead" THEN IF [Country] = "United States" THEN TRUE ELSE FALSE END WHEN "International_Lead" THEN IF [Country] != "United States" AND NOT ISNULL([Country]) THEN TRUE ELSE FALSE END ELSE END



2. Click OK.

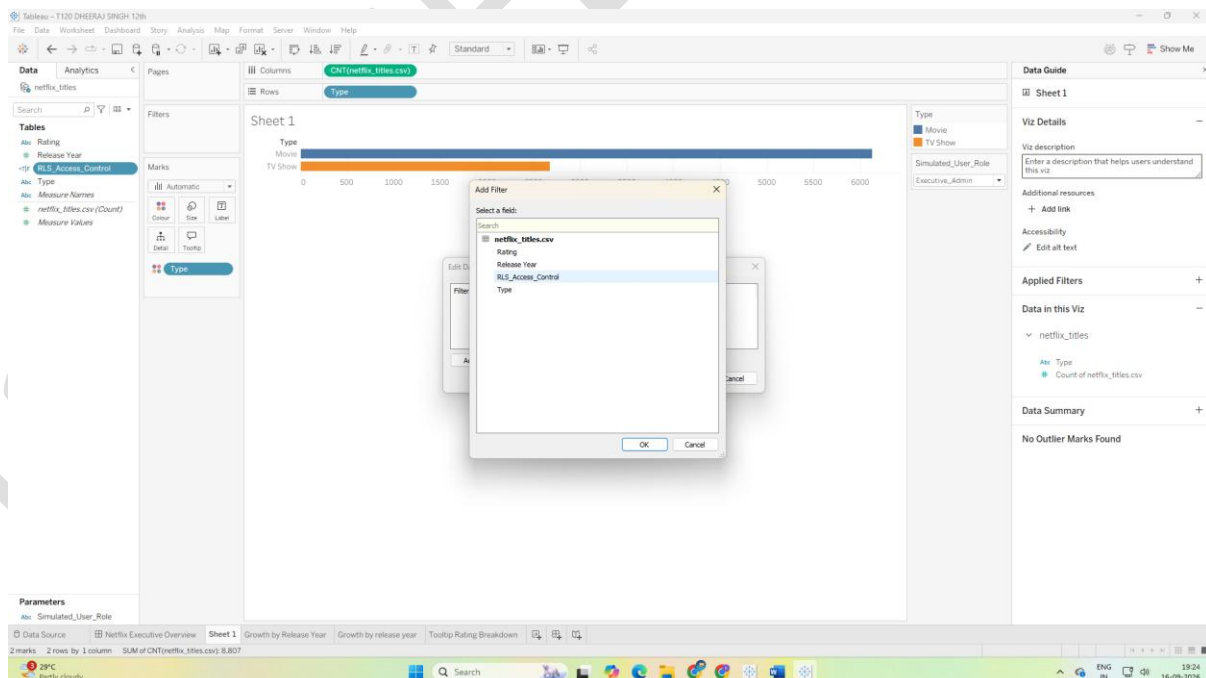
Step 3: Enforce Access Control at the Data Ingestion Layer

1. Go to the top application menu bar --> select Data --> hover over netflix_titles --> select Edit Data Source Filters....

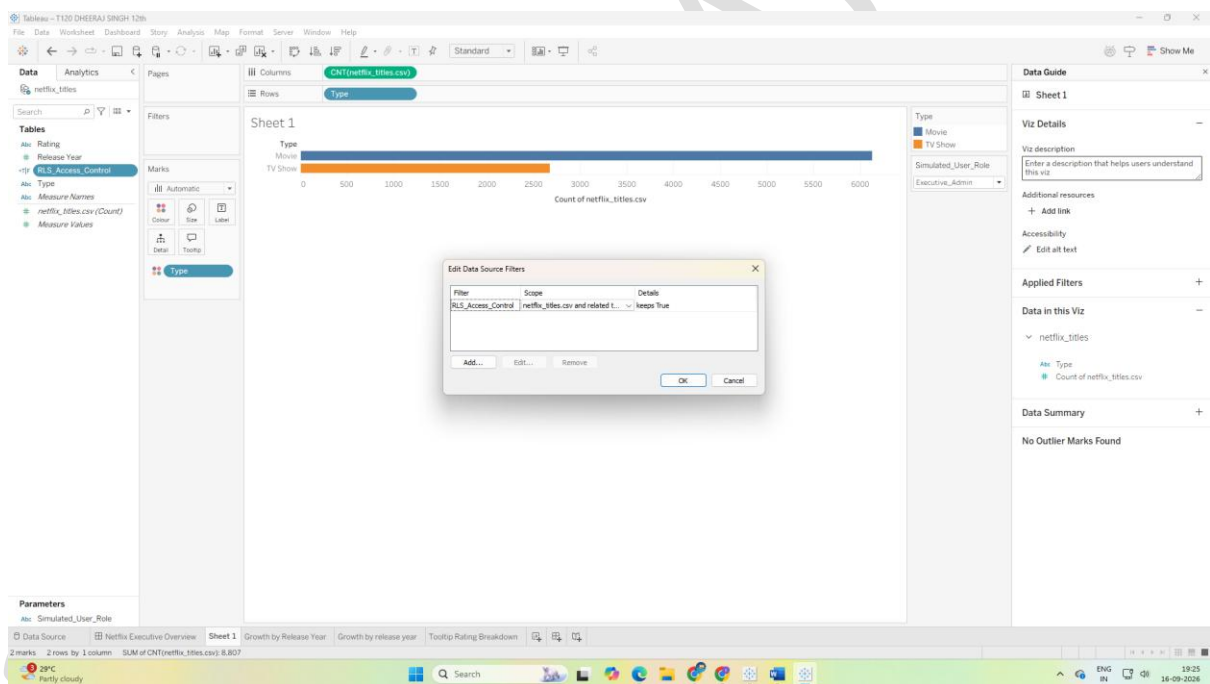
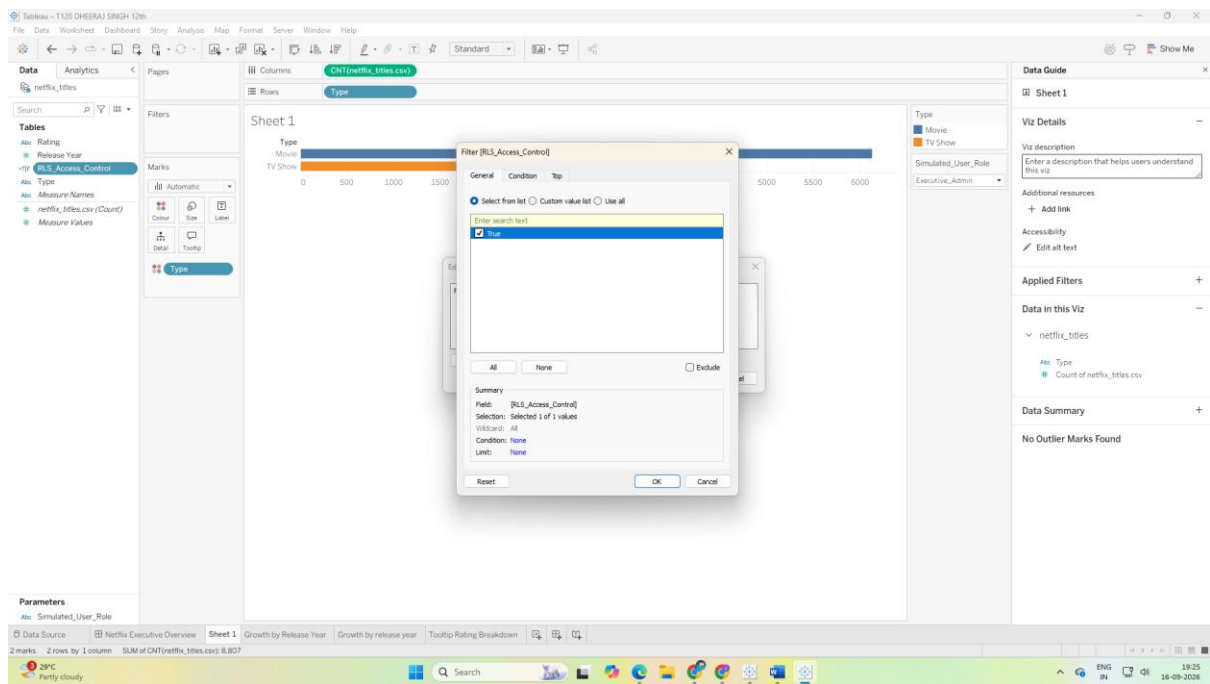


2. In the Data Source Filters dialog, click Add....

3. Select RLS_Access_Control from the field list --> click OK.



4. In the filter dialog, check the box for True --> click OK, then click OK again.



Step 4: Test and Verify Role-Based Permissions

Switch the Simulated_User_Role parameter dropdown on your dashboard to observe realtime access enforcement:

- Select Executive_Admin: The entire global catalog of 8,807 titles across all countries is accessible.
- Select US_Regional_Lead: The dashboard automatically filters to show only the 2,818 United States titles.
- Select International_Lead: All US titles are masked, leaving only non-US international content.

- Select Restricted_Guest: The access filter returns FALSE for all rows, blanking out the view entirely (zero data exposure).

